

Targeting Cell Senescence and Senolytics: Novel Interventions for Age-Related Endocrine Dysfunction

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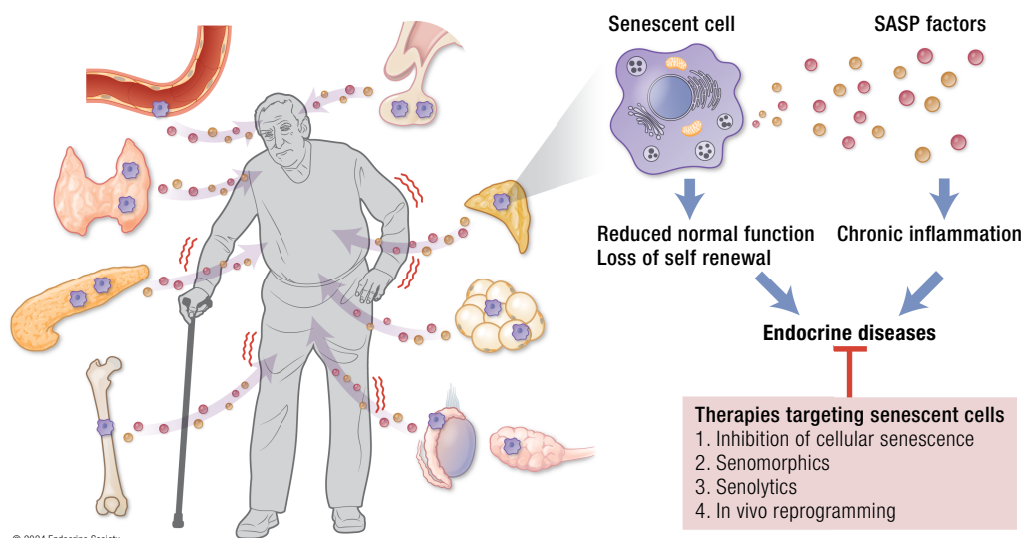
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Abstract

Multiple changes occur in hormonal regulation with aging and across various endocrine organs. These changes are associated with multiple age-related disorders and diseases. A better understanding of responsible underlying biological mechanisms could help in the management of multiple endocrine disorders over and above hormone replacement therapy (HRT). Cellular senescence is involved in multiple biological aging processes and pathologies common in elderly individuals. Cellular senescence, which occurs in many older individuals but also across the lifespan in association with tissue damage, acute and chronic diseases, certain drugs, and genetic syndromes, may contribute to such endocrine disorders as osteoporosis, metabolic syndrome, and type 2 diabetes mellitus. Drugs that selectively induce senescent cell removal, “senolytics,” and drugs that attenuate the tissue-destructive secretory state of certain senescent cells, “senomorphics,” appear to delay the onset of or alleviate multiple diseases, including but not limited to endocrine disorders such as diabetes, complications of obesity, age-related osteoporosis, and cancers as well as atherosclerosis, chronic kidney disease, neurodegenerative disorders, and many others. More than 30 clinical trials of senolytic and senomorphic agents have already been completed, are underway, or are planned for a variety of indications. Targeting senescent cells is a novel strategy that is distinct from conventional therapies such as HRT, and thus might address unmet medical needs and can potentially amplify effects of established endocrine drug regimens, perhaps allowing for dose decreases and reducing side effects.

Graphical abstract



Key Words: cellular senescence, endocrine diseases, diabetes, osteoporosis, SASP, senomorphics, senolytics

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Abbreviations: 53BP1, Tumor suppressor p53-binding protein 1; γ H2AX, γ phosphorylated form of the histone H2AX; ACP, adamantinomatous craniopharyngioma; AFFIRM, Alleviation by Fisetin of Frailty, Inflammation, and Related Measures; Akt, protein kinase B; Ang II, angiotensin II; ApoE, apolipoprotein E; ATC, anaplastic thyroid cancer; ATTAC, apoptosis through targeted activation of caspase 8; BCL-xL, B-cell lymphoma-extra large; BPTES, bis-2-(5-phenyl- acetamido-13,4-thiadiazol-2-yl) ethyl sulfide; CAR-T, chimeric antigen receptor T cell; CDK, cyclin dependent kinase; CDKI, cyclin-dependent kinase inhibitor; CTL, cytotoxic T lymphocyte; Cxcl1, chemokine ligand 1; Cxcl2, chemokine ligand 2; D, dasatinib; D2, deiodinase; DHEA, dehydroepiandrosterone; EC, endothelial cell; EFNB1, ephrin B1; EV, extracellular vesicle; FOXO, forkhead box O; GH, growth hormone; GPNMB, glycoprotein nonmetastatic melanoma protein B; HGPS, Hutchinson-Gilford progeria syndrome; HRT, hormone replacement therapy; IGF-1, insulin-like growth factor 1; IGFBP, insulin-like growth factor-binding protein; IL- α , interleukin- α ; IL-6, interleukin-6; IL-8, interleukin-8; IRS2, insulin receptor substrate 2; JAK, Janus kinase; KCNJ5, potassium inwardly rectifying channel subfamily J member 5; KGA, kidney-type glutaminase; Ki-67, marker of proliferation Ki-67; KO, knockout; MAPK, mitogen-activated protein kinase; miRNA, microRNA; MMP, matrix metalloproteinase; mtDNA, mitochondrial DNA; mTOR, mammalian target of rapamycin; mTORC1, mammalian target of rapamycin complex 1; NAD, nicotinamide adenine dinucleotide; NAFLD, nonalcoholic fatty liver disease; NASH, nonalcoholic steatohepatitis; NF, nuclear factor; NK, natural killer; OIS, oncogene-induced senescence; OSKM, Oct4, Sox2, Klf4, and c-Myc; p16^{INK4a}, cyclin-dependent kinase inhibitor 2A; p19^{INK4d}, cyclin-dependent kinase 4 inhibitor D; p21^{CIP1/WAF1}, cyclin-dependent kinase inhibitor 1 (Cdkn1a); p38MAPK, p38 mitogen-activated protein kinase; p53, cellular tumor antigen p53; 53BP1, p53-binding protein 1; PAI-1, plasminogen activator inhibitor-1; PAI-2, plasminogen activator inhibitor-2; PAPP-A, pregnancy-associated plasma protein A; PI3K, phosphoinositide 3-kinase; PI3KCD, phosphatidylinositol-4,5-bisphosphate 3-kinase delta catalytic subunit; Q, quercetin; Rb, retinoblastoma protein; ROS, reactive oxygen species; SADS, senescence-associated distention of satellites; SASP, senescence-associated secretory phenotype; SA- β gal, senescence-associated β -galactosidase; SCAPs, senescent cell antiapoptotic pathways; siRNA, small interfering RNA; STAT JAK, signal transducer and activator of transcription; T2DM, type 2 diabetes mellitus; T3, triiodothyronine; T4, thyroxine; TAFs, telomere-associated foci; TAME, targeting aging with metformin; TH, thyroid hormone; Timp1, TIMP metalloproteinase inhibitor 1; TIS, therapy-induced senescent cells; TSH, thyrotropin; TNF α , tumor necrosis factor α ; uPAR, urokinase-type plasminogen activator receptor.

ESSENTIAL POINTS

- Endocrine organs accumulate senescent cells with increasing age, causing functional decline
- Cellular senescence is a crucial antitumor mechanism, but senescent cell accumulation is detrimental to tissue function
- Senescent cells produce proinflammatory, profibrotic signals dubbed the “senescence associated secretory phenotype”
- Senescent cells can be selectively removed by so-called senolytic drugs, silenced by senomorphic drugs, or reprogrammed back to a nonsenescent state
- Initial results from senolytic clinical trials are promising, but these treatments are new and further studies are warranted, not the least to investigate for potential long-term side effects
- In vivo models targeting senescent cells have been shown to alleviate several endocrine disorders, including diabetes and osteoporosis
- Antisenescence therapies may be an effective adjunct to existing treatments for endocrine disorders, including but not limited to diabetes mellitus, osteoporosis, infertility, adrenal gland dysfunction, and hypothyroidism

Countries across the world are experiencing growth both in numbers and the proportion of older individuals. By 2050, the world's population of people aged 60 years and older will double (to 2.1 billion) and those aged 80 years or older are expected to triple (to 426 million) (1). Endocrine disorders such as obesity, diabetes mellitus, osteoporosis, and gonadal and thyroid dysfunction are major causes of morbidity and mortality and contribute to an increasingly large health care burden, especially in older individuals (2). In addition, an aging endocrine system contributes to age-associated diseases and geriatric syndromes (3). Associations between increasing age and endocrine function are manifested by declines of several blood hormone levels, such as estrogens (in women), testosterone (in men) (4-6), growth hormone (GH) (7-10),

aldosterone (11), and melatonin (12, 13), changes that may correlate with the development of diseases and geriatric syndromes in older patients. For example, low postmenopausal estrogen levels are associated with osteoporosis as well as cardiovascular and cerebrovascular diseases (14, 15), insufficient GH secretion or effects may contribute to loss of skeletal muscle and increased adiposity, decreases in circulating aldosterone to lightheadedness and orthostatic hypotension, and reduced melatonin production to disrupted sleep-wake cycles (ie, the circadian rhythm). On the other hand, the levels of some hormones, such as cortisol (16-18) and thyroid hormones (THs) (19-22), exhibit little negative or positive correlation with increased age. Even when hormone levels are normal or increased, the endocrine system may not function effectively due to changes in target organ function and hormonal responsiveness. For example, insulin resistance and hyperinsulinemia are often present in older individuals (23). Hormone replacement therapy (HRT) has been used to counteract some of these age-related changes in hormone levels or sensitivity (24), although use of HRT can have both benefits and risks (25, 26). How increasing age affects circulating hormone levels requires further study, as new research may disrupt previously held dogmas, indicating the difficulty of drawing generalized conclusions based on studies in younger populations regarding hormone levels in older individuals (20). Recent findings indicate that the association between androgens and aging is not clear. For example, after controlling for comorbidities (27) and aldosterone secretion may increase slightly with aging (28). The complex regulatory systems involved in hormone secretion and activity are influenced by such factors as genetic background, race, ethnicity, sex, lifestyle choices, metabolic factors, and concurrent medical conditions (20). A better understanding about the other biological changes that occur in the endocrine system with aging beyond hormone secretion and processing might help in developing interventions to extend health span, the period of life free of substantial disabilities, chronic disorders and diseases, and decreased physical reserve and resilience.

The effects of cellular senescence and senescent cell accumulation on biological aging processes are becoming increasingly better understood. There is in vitro and in vivo evidence suggesting senescent cell accumulation can contribute to endocrine dysfunction and endocrine disorders such as in the

pancreas in type 2 diabetes mellitus (T2DM) (29), adipose tissue in diabetes, obesity, and advanced age (30–37), the kidneys in diabetic nephropathy and obesity (38, 39), the liver in association with hyperinsulinemia, metabolic syndrome, and age-dependent steatosis (40–42), age-related and radiation-induced osteoporosis (43–46), endometrial and uterine dysfunction (47), cancers and their complications, as well as side effects of cancer treatments on endocrine function (48, 49), cardiovascular diseases, including those linked to high-fat diets (50–52), and anxiety associated with obesity (53), among other conditions.

The term “cellular senescence” originates from the 1960s, when Hayflick and Moorehead demonstrated that human cultured fibroblasts have limited replicative potential (54). After exhausting their replicative capacity, senescent cells remain metabolically active, and can persist in tissues, causing dysfunction. Apart from not contributing to tissue growth and dampening tissue regeneration due to their cell cycle arrest, senescent cells can release proinflammatory, profibrotic, and proapoptotic signals as well as progrowth factors and factors that modulate progenitor function as part of the senescence-associated secretory phenotype (SASP) (55–58). The proteins, peptides, bioactive small molecules, and non-coding nucleotides that can comprise the SASP are highly variable: The composition of secreted signals depends on the type of cell that became senescent, the trigger that induced senescence, the microenvironment, and time since induction of senescence (59–64).

Targeting and removing senescent cells or reducing the release of certain SASP factors in age-associated diseases have the potential to yield new options for treating diseases without any established effective interventions to date. These novel agents might also be eventually used as adjuvants to HRTs or other therapies, allowing for dose reductions, potentially reducing side effects. Targeting fundamental aging mechanisms might delay, prevent, alleviate, or treat multiple disorders and diseases and complex conditions in older patients including frailty, immobility, sarcopenia/muscle wasting, urinary incontinence, falls, mild cognitive impairment, increased risk for delirium, chronic skin ulcers, loss of resilience against stressors such as surgery and infections, or diminished responses to immunogenic stimuli such as vaccinations (65–69). Since these adverse geriatric conditions are encountered frequently in individuals with diabetes, osteoporosis, and other endocrine conditions (70–72), root-cause treatments targeting fundamental aging mechanisms might be particularly beneficial for such patients.

Markers of Cellular Senescence

Senescent cells have an enlarged and flattened morphology, frequently have high p53/p21^{CIP1/WAF1} and/or p16^{INK4a}/Rb protein levels, and have DNA damage foci (particularly in telomeres) (Fig. 1). Another biomarker for detecting senescent cells in culture or tissue samples is the lysosomal enzyme, senescence-associated- β -galactosidase (SA- β gal), with enzymic activity at pH 6.0 as opposed to lysosomal enzymes from non-senescent cells (73). However, this marker is not highly sensitive or specific. A central feature of senescence is replicative arrest and lack of DNA replication. This can be detected using clonal plate dilution assays or by the absence of nucleoside analogue incorporation (eg, 5-bromodeoxyuridine or [3H] thymidine) (74). Immunostaining for proliferation

markers, such as proliferating cell nuclear antigen and the marker of proliferation, Ki-67 (Ki-67), can also help in detecting senescent cells. Increased proinflammatory SASP factors, including interleukin-6 (IL-6), interleukin-1 α (IL-1 α), interleukin-8 (IL-8), monocyte chemoattractant protein 1, plasminogen-activated inhibitor 1 (PAI-1), plasminogen-activated inhibitor 2 (PAI-2), and matrix metalloproteinases (MMPs) can indicate increased senescent cell burden and can be analyzed in blood, tissues, or cells (64). However, not all senescent cells are proinflammatory and proapoptotic (63, 75). Hence, there are many different senescence markers with varying degrees of specificity and sensitivity. As of today, there is no single marker for accurately measuring senescent cell accumulation, and the establishment of new cellular senescence markers or composite scores comprising key markers is a pressing issue. Ideally, these markers or scores should be detectable in noninvasively collected samples such as blood and urine.

Senescent cells generally have persistent DNA damage. Nuclear senescence-associated heterochromatin foci can be used to identify senescence induced by activated oncogenes such as H-RAS and BRAF and stressors that impede DNA replication (76, 77). Increased activity of p38 mitogen-activated protein kinase (p38MAPK) or the γ phosphorylated form of the histone H2AX (γ H2AX) reflects activated DNA damage responses, which along with depleted or irreparably damaged telomeres, can indicate cellular senescence. DNA damage response factors that persist at sites of damage and nuclear foci, which can be detected cytologically, can also serve as indicators of cellular senescence. Telomere-associated DNA damage foci (TAFs) that accumulate within telomeric sequences, as well as colocalization of γ H2AX and p53-binding protein 1 (53BP1) with telomeres, are also indicators of the senescent state. Pericentromeric satellite heterochromatin undergoes decondensation in senescent cells, leading to senescence-associated distention of satellites (SADs). SADs appear earlier and more consistently than heterochromatin foci, reflecting an early and potentially key event in cellular senescence (76, 78). Damage-associated molecular pattern factors, such as high-mobility group box 1 localization or molecules released by stressed cells undergoing cell death, such as mitochondrial DNA (mtDNA), reflect cellular damage and can indicate increased senescent cell abundance (79).

The enzyme α -Klotho regulates multiple endocrine processes, such as insulin-like growth factor 1 (IGF-1) signaling (80), while also modulating mammalian target of rapamycin (mTOR) (81, 82), cyclic adenosine monophosphate (83), p53/p21^{CIP1/WAF1} (84), and Wnt protein levels (85). It is also involved in mineral metabolism, contributing to phosphate homeostasis. α -Klotho has come to light as a geroprotective factor that protects against physiological stresses such as oxidative damage and hypoxia. α -Klotho is also protective against the side effects of cytotoxic drugs. It is secreted by the distal renal tubule and so is present in urine. Importantly, α -Klotho is inversely and causally linked to senescent cell burden, and senolytics increase urinary α -Klotho in humans (86). This suggests that α -Klotho, along with other measures, could be a useful “gerodiagnostic” marker with respect to senescent cell abundance and other fundamental aging processes.

Analysis of microRNAs (miRNAs) that may be specific to senescent cells (87, 88), other nucleotides such as cell-free mtDNA (79), senescence-specific epigenetic profiles, and

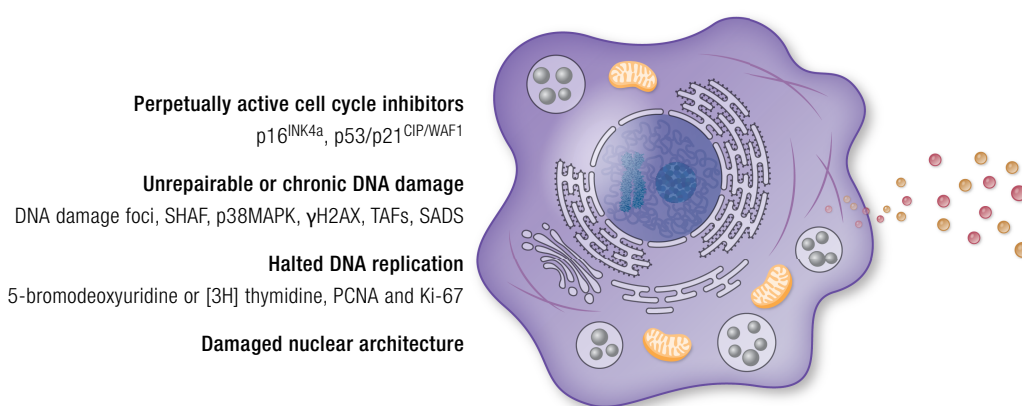


Figure 1. Senescent cell markers. Several senescence markers have been identified based on the molecular biology of senescent cells. Relatively new markers used in recent years are senescence-associated heterochromatin foci, telomere-associated foci, and senescence-associated distention of s-atellites. However, many of these senescence markers are nonspecific due to the heterogeneity of senescent cells, and to date, no single marker is a fully sensitive and specific indicator, especially for clinical use. Improvements have been made by developing composite scores or signatures comprising multiple markers. Refinement of indicators of senescent cell burden in vivo for clinical application, especially those that can be assayed noninvasively, is needed.

senescence-associated small extracellular vesicles (EVs) including exosomes (89, 90), microsomes, or mitosomes could also be viable strategies for assessing senescent cell burden. Progress has been made with miRNAs, short (20–24 nt) non-coding RNAs that are involved in posttranscriptional regulation of gene expression. Several miRNAs that are differentially expressed with aging and by senescent cells have been reported (87, 91, 92).

Since none of these markers can be used as a reliable senescence biomarker on its own, combinations of such indicators may be a more reliable reflection of senescent cell burden (74, 93, 94). Such composite scores could be useful for reflecting senescent cell abundance and following therapeutic efficacy in clinical trials of agents targeting senescent cells (95, 96). Cell cycle arrest proteins (eg, p53, p21^{CIP1/WAF1} and p16^{INK4a}) can be measured in body fluids. Among these, p16^{INK4a} expression in peripheral blood T cells has been used to estimate senescent cell burden (97, 98). Biopsy samples of, for example, adipose tissue or skin could be an alternative for assessing senescent cell burden, potentially along with imaging modalities.

Senescent Cells Can Accumulate With Age and Drive Morbidity

Accumulating evidence from in vitro and in vivo studies using the senescent markers mentioned earlier indicate that senescent cells accumulate in multiple endocrine organs with aging, especially in older individuals with impaired function, decreased physical resilience, and/or multimorbidity. Senescent cell accumulation may contribute to the onset and progression of several endocrine diseases (Fig. 2).

Diabetes Mellitus, Metabolic Syndrome, and Senescence

The prevalence of T2DM increases with age (99, 100), together with the hallmarks of this condition such as increased insulin levels and peripheral insulin resistance. Adipose tissue is a metabolically dynamic organ that is the primary site of excess energy storage, but it also serves as an endocrine organ

capable of synthesizing a number of biologically active compounds that regulate metabolic homeostasis, such as tumor necrosis factor- α (TNF α), IL-6, IL-8, leptin, adiponectin, angiotensin, resistin, and PAI-1 (101–103). Two decades ago, it was found that aging is linked to increased adipose tissue inflammation and decreased capacity of cloned adipose progenitors and mesenchymal stem cells for differentiation and replication. This decline in adipogenesis correlated with insulin resistance (104). These findings suggest that presenescent and senescent cells accumulate in adipose tissue with aging since impaired replication is a hallmark of cellular senescence. It was then found that insulin resistance correlates with increased markers of cellular senescence in fat tissue, including adipose tissue β -galactosidase, which is a senescence-linked marker of increased lysosomal activity, as well as increased levels of PAI-1, p53, and cyclin D kinase inhibitors, including p16^{INK4a} (37, 105, 106). Excessive caloric intake leading to insulin resistance increases SA- β gal activity with increased p53 and p21^{CIP1/WAF1} expression in adipose tissue compared to normal caloric intake (105). These changes were related to accumulation of reactive oxygen species (ROS), which can drive senescence or be a product of senescent cells (105). In adipose tissue-specific p53 knockout (KO) mice, insulin resistance caused by a high-fat diet was significantly attenuated, and senescence markers in adipose tissue were decreased even during increased caloric intake (105). These findings suggest that insulin resistance linked to obesity can be mediated by cellular senescence in adipose tissue. Complications of insulin resistance, including hepatic steatosis progressing to non-alcoholic fatty liver disease (NAFLD) and nonalcoholic sclerosing hepatitis (NASH), diabetic/high-fat diet-induced kidney disease, diabetic cardiovascular dysfunction, macular degeneration, and obesity-related neuropsychiatric dysfunction are associated with senescent cell accumulation in the affected organs (34, 38–42, 52, 107, 108).

Insulin/Insulin-like Growth Factor-1 Signaling and Aging

Examples of hormones affected by aging include insulin and IGF-1, which regulate metabolic balance, anabolic activity,

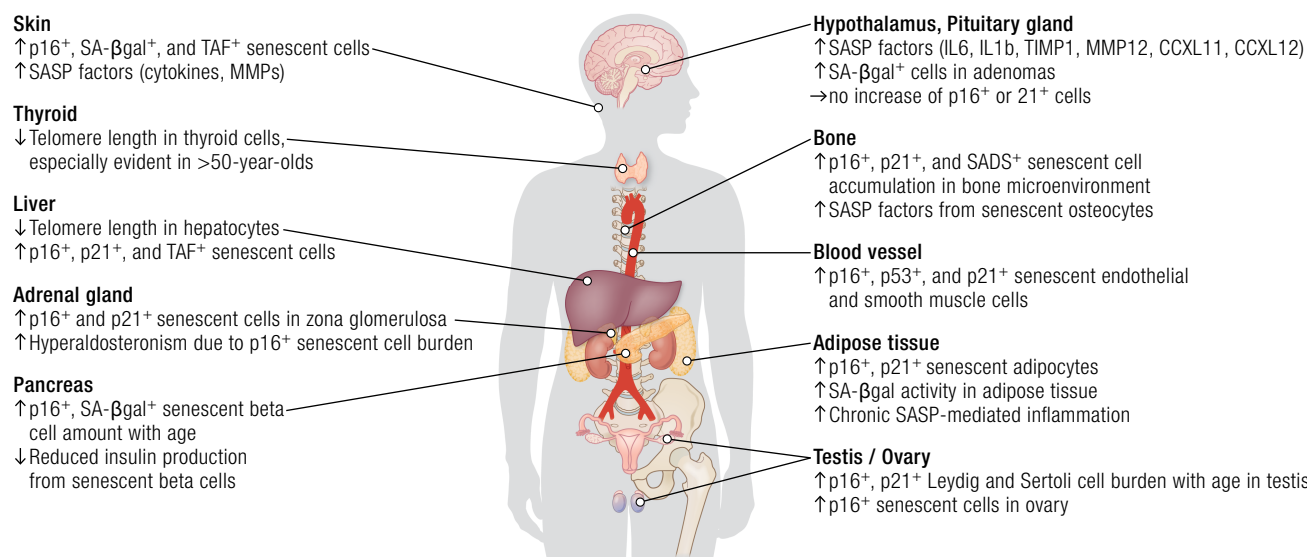


Figure 2. Senescent cell accumulation in endocrine tissues with aging. With increasing age, senescent cells can accumulate in tissues, as demonstrated by assays of p16^{INK4a}/p21^{WAF1/CIP1} expression, increased senescence-associated β-galactosidase activity, or shortened telomeres. Another approach is to monitor levels of senescence-associated secretory phenotype factors. Senescent cells can disrupt tissue homeostasis in endocrine and other organs and dysregulate hormone production and target organ effects, contributing to worsening health outcomes.

and replication and differentiation of multiple types of cells (109). In 1997, it was reported that a mutation leading to decreased expression of *daf-2* (Dauer formation-2), which encodes an insulin/IGF-1 receptor in the nematode *Caenorhabditis elegans*, results in a 2- to 3-fold increase in lifespan (110, 111). This gene is conserved across species ranging from yeast to mammals (112). Other genes involved in insulin/IGF-1 signaling have also been shown to extend lifespan when suppressed. Examples include the transcription factor, *daf-16*, a *C. elegans* homologue of the forkhead box O (FOXO) gene in humans that is downstream in the IGF-1/Akt signaling pathway (113, 114), and *Sch9*, a gene homologous to protein kinase B (Akt) (115, 116). In *Drosophila*, mutations leading to decreased expression of the insulin/IGF-1 receptor or in insulin receptor substrate (IRS) 2-like molecules have also been shown to be associated with prolonged lifespan (117). Increases in lifespan from reduced insulin signaling are conserved in more complex organisms. In mammals, in which insulin and IGF-1 have separate receptors, heterozygous IGF-1 KO mice live an average of 30% longer than controls (118) and mice lacking adipose tissue insulin receptors live 18% longer, as do mice lacking brain IRS2 (119, 120). Mice with genetic GH deficiency or humans with isolated GH deficiency are protected from aging phenotypes and show longevity (121). The National Institute on Aging's Primate Aging Research Project found that caloric restriction extends lifespan in primates. Caloric restriction of rhesus macaques over a 20-year period resulted in animals that were biologically "younger," with shinier hair and less likelihood to develop age-related diseases (122). These monkeys also had low circulating insulin (123, 124). These findings suggest that insulin/IGF-1/Akt signaling might have a role in primate aging (122). In the Baltimore Longitudinal Study of Aging, which began in 1958, approximately 700 older men were followed for 25 years. Low insulin levels were associated with longevity (125), further suggesting that insulin/IGF-1/Akt signaling is involved in the development of age-related dysfunction and diseases (126, 127).

Consistent with our Unitary Theory of Fundamental Aging Mechanisms (128, 129), age-related changes in IGF-1 and senescent cell abundance appear to be interlinked (130). The zinc metalloproteinase plasma protein-A (PAPP-A) increases IGF-1 bioavailability by cleaving insulin-like growth factor binding proteins (IGFBPs), particularly IGFBP-4 (131). IGFBPs bind IGF-1, hindering receptor activation. After cleavage of IGFBP-4 by PAPP-A, IGF is liberated from the binding protein and IGF signaling is initiated in the pericellular environment. Inhibiting PAPP-A activity by gene deletion, and thereby decreasing IGF-1 signaling, appears to extend lifespan by up to approximately 30% in naturally aging mice (132). This was also observed in PAPP-A KO mice on a high-fat diet, and even in already adult mice after PAPP-A gene expression was knocked-down (132-134). In the conditioned medium from senescent (senescence induced by etoposide) adult human primary preadipocytes, PAPP-A proteolytic activity was more than 12-fold higher compared to conditioned medium from nonsenescent cells, indicating that PAPP-A is an SASP component. Proteolytically active PAPP-A was also abundant on the surface of EVs secreted by senescent preadipocytes. These findings link increased IGF-1 activity to cellular senescence (130).

Perhaps paradoxically, despite increases in lifespan related to low IGF-1 levels, low levels of circulating IGF-1 in plasma have also been linked to low muscle mass, a key component of frailty (135), although this has not been found consistently and requires further study. Frailty indices include grip strength, gait speed, or other indirect measurements of muscle mass, strength, and function. IGF-1, along with its systemic anabolic effects, directly enhances muscle protein synthesis, driving hypertrophy (136), as well as increasing the abundance of the satellite cells that surround and support skeletal muscle (137). IGF-1, through Akt, increases mTOR activity, which in turn increases protein synthesis (138). It also decreases FOXO levels and thus protein breakdown (139). In a 1990 study of men older than 60 years with low IGF-1 levels, exogenous GH administration increased bone density and

lean muscle mass while decreasing adipose tissue mass (140). However, studies with recombinant human GH and GH secretagogues failed to demonstrate benefits that outweigh risks such as increases in insulin resistance linked to weight gain. Resistance training may decrease frailty and improve health in older patients (141–144) and stimulates the hypothalamic-pituitary GH-IGF-1 axis through stimuli from muscles (145). Thus, relationships among IGF-1, muscle function, frailty, health span, and lifespan are complex.

Osteoporosis and Senescence

With increases in the number of older people, the prevalence of osteoporosis has been increasing. Reduction in bone quality brings with it reduced quality of life, mainly due to pathological fractures (146, 147). The pathophysiology of osteoporosis is multifactorial, and vitamin D deficiency is a risk factor. With age, the capacity of the skin to synthesize vitamin D decreases (148) and senescent cells accumulate (149), but whether this association is causal remains to be determined.

Detrimental effects of senescent cell accumulation in bone have recently come to light. The senescence biomarkers, p16^{INK4a} and p21^{CIP1/WAF1} were elevated in iliac crest needle biopsies from older postmenopausal compared to younger premenopausal women. Despite the heterogeneous nature of the biopsy samples, including variations in cellular composition and fat abundance, SASP factors were also increased in biopsies from the older participants (78). Accumulation of senescent cells in the bone microenvironment is linked to increased bone resorption by osteoclasts and reduced bone formation by osteoblasts, leading to reduced bone density (150). The senescent osteocytes that accumulate with aging in mice have increased expression of multiple SASP factors compared to young mice, as well as age-associated upregulation of SASP factor production in bone marrow myeloid cells. These findings suggest that senescent osteocytes and their SASP may contribute to age-related bone loss and that their removal may be a therapeutic strategy for age-related (as opposed to postmenopausal) osteoporosis (151).

Hypothalamus and Pituitary Gland

The hypothalamus and the pituitary gland are key regulatory organs of the endocrine system, and this regulatory function can become disrupted with aging. Age-related loss of hypothalamic regulation might be linked to SASP factors, as hypothalamic pro-opiomelanocortin neurons were increased by rapamycin, a known SASP inhibitor (152). In rodents, the hypothalamus becomes less sensitive to several feedback processes with aging (153, 154). Hypothalamic arcuate nucleus GH-releasing hormone secretion is decreased in older individuals while paraventricular nucleus somatostatin secretion is increased, contributing to decreased GH levels related to reduced frequency and lower amplitude of secretory pulses (155–157). In a recent study, age-related changes were detected in the hypothalamic expression of the SASP factors IL-6, IL-1 β , TIMP metalloproteinase inhibitor 1 (Timp1), Mmp12, Cxcl1, and Cxcl2. However, the senescent cell markers p16^{INK4a} and p21^{Cip1/Waf1} were not increased (64). This needs to be explored further at the single-cell level in the hypothalamus because in some tissues, for example, muscle, it was recently discovered that a very small, difficult to detect compartment of senescent cells contributes to substantial dysfunction (158). Deleterious effects of this small

senescent cell fraction on muscle function were dramatically alleviated by senolytics.

The diurnal rhythmic release of melatonin is regulated, in part, by a circadian clock located in the suprachiasmatic nucleus of the hypothalamus. This release can become dysregulated with increasing age, leading to decreased levels of melatonin and poor sleep quality (159). In mice, disruption of circadian clock genes such as *Bmal1* or *Clock* reduced lifespan (160–162) and accelerated development of age-related phenotypes. Interestingly, poor sleep quality may be correlated with increased senescent cell burden, evidenced in humans by increased numbers of circulating senescent T cells (163) and increased p16^{INK4a} gene expressing peripheral blood mononuclear cells (164). In addition, senescent cells can accumulate in the aorta after induced sleep fragmentation (165). More work needs to be conducted to evaluate the connection between sleep disturbances and senescent cells, perhaps focusing on effects of hypothalamic senescent cell burden.

The pituitary gland is prone to adenoma formation with aging, with an estimated prevalence of clinically silent adenomas in older individuals of up to 20% (166). Increased SA- β gal activity has been found in pituitary adenomas compared with normal tissue, possibly due to oncogene-induced senescence (OIS) (167, 168). In this instance, the appearance of senescent cells may be beneficial, since OIS can slow tumor growth. Perhaps this could be the case for other endocrine tumors, but further research into this is needed.

Gonadal Dysfunction and Sex Hormone Disorders

With increasing age, the ovary and uterus in females and the testis in males become dysfunctional (169), with decreased sexual function, infertility, sleep, and mood disturbances, and loss of muscle mass (170). Gonadal dysfunction with aging may be related to cellular senescence and mitochondrial dysfunction (171), and senolytic or senomorphic interventions may have the potential to delay menopause and extend the female reproductive window (172). In aged rats, p16^{INK4a} levels increase in the ovaries and testes (173). In aged compared to young dogs, a 4-fold increase in p21^{CIP1/WAF1} expression in testicular fibroblasts and 8 times more senescent Leydig cells in the testes were found (174). Senescent cells may interfere with the differentiation of endometrial stromal cells into decidual cells, impeding embryonic implantation and placentation, leading to infertility (175). However, a recent study using p16^{INK4a}-KO mice found no improvement of infertility in an alkylating agent-induced primary ovarian insufficiency model on inhibiting p16^{INK4a}-induced senescence (176). More work needs to be performed to determine if the accumulation of senescent cells in gonadal tissues causes dysfunction.

Adrenal Gland

The adrenals secrete cortisone, aldosterone, and dehydroepiandrosterone (DHEA), among other hormones. The connection between DHEA and longevity has been investigated in several epidemiological studies (177–179). In the previously mentioned Baltimore Longitudinal Study, high DHEA levels were associated with longevity (125). Dysregulated function of the adrenals can have wide-reaching consequences, even affecting brain function and mood. The daily cortisol rhythm is disrupted in patients with dementia, along with decreased

DHEA levels. High levels of circulating cortisol may induce apoptosis of hippocampal neurons, while high levels of DHEA protect them (180, 181). Senescent cell accumulation in the adrenals has been suggested to have detrimental effects, with senescent cells in the zona glomerulosa being linked to hyperaldosteronism (182, 183). In a recent paper investigating potassium inwardly rectifying channel subfamily J member 5 (KCNJ5)-mutated aldosterone-producing adenomas, p21-induced cell cycle arrest was correlated with higher aldosterone-to-renin ratios (184). In KCNJ5-mutated and wild-type aldosterone-producing adenomas, compact tumor cells were more likely to be senescent than intratumor clear cells (184). More work is needed to test if and how cellular senescence affects adrenal function and tumor formation. Since DHEA replacement therapy in older women did not lead to major benefits (177, 185), widespread use of DHEA supplementation as a gerotherapeutic intervention is not supported as of today.

Thyroid and Parathyroid Glands

Studies in human and animal models have suggested an inverse relation between TH levels and longevity. Older men appear to have decreased sensitivity to thyrotropes, possibly related to decreased thyrotropin-releasing hormone 24-hour rhythmicity or increased somatostatin (186, 187). In vivo studies in several long-lived small mammalian species suggest that lower TH levels are associated with extended longevity, such as a recent study in which 4 small mammalian species were studied (188). This indicated there is an inverse relation between thyroxine (T4) levels and species maximum lifespan. Increased mitochondrial activity related to TH may link increased TH to accelerated development of aging phenotypes, perhaps related to increased generation of DNA-damaging ROS. This appears to be associated with the accelerated appearance of aging changes in vitro and in vivo (189). ROS and cellular senescence are causally linked. ROS can induce cells to become senescent, and senescent cells activate macrophage degradation of nicotinamide adenine dinucleotide (NAD), which then leads to increased ROS generation, especially by those innate immune cells that can induce other types of nonsenescent cells to become senescent (190, 191). Studies are needed to determine if these interlinked cellular-senescence-mediated processes are accentuated by TH. Also pointing to potential links between senescence and the effects of thyroid function on the progression of aging phenotypes is the observation that caloric restriction, which can delay the development of aging phenotypes, both reduces circulating TH and decreases senescent cell burden (173). Progressive telomere shortening, which can be both a cause and biomarker of cellular senescence, has been noted in the thyroid and parathyroid glands of humans. Thyroid and parathyroid telomeric erosion becomes evident in people older than 50 years, which is later than for other tissues in which telomere shortening has been reported (192). Perhaps this is due to the slow turnover of thyroid cells. Older women tend to have higher thyrotropin (TSH) levels than younger women and, in both sexes, triiodothyronine (T3) tends to be lower and circulating antithyroid antibodies higher in older than younger populations (193). In geographic areas where iodine intake is high, TSH levels tend to increase with age, whereas in areas with lower iodine intake, circulating TSH levels generally decrease with increasing age (193, 194). Long-term residency in areas

with high iodine content in the drinking water has been associated with increased longevity (195-197). Although epidemiological data suggest a relationship between iodine intake and longevity, little is currently known about whether this is an indirect effect of changes in TH or iodine homeostasis.

The most aggressive thyroid neoplasm is anaplastic thyroid cancer (ATC). Historically, the mean survival time after this diagnosis has been established is 4 months (198). In the thyroid gland, type 2 deiodinase (D2) converts T4 into the more metabolically active T3, which is crucial for ATC cell proliferation. Interestingly, treatment by the D2 inhibitor, reverse T3, induces cellular senescence in these tumor cells (184). Together with senolytic therapy to remove these now-senescent cancer cells in a 2-step process ("1:2 punch" approach) (199, 200), this may become a valid treatment strategy for patients suffering from this type of cancer.

Vitamin D Metabolism

The conversion of the vitamin D precursor, 7-dehydrocholesterol, into previtamin D₃ occurs in the skin with exposure to ultraviolet radiation in sunlight. The capacity of the skin to synthesize previtamin D₃ decreases by up to 50% with increased age (148, 201, 202). Cellular senescence plays a major role in skin aging, with senescent cells accumulating that produce collagenases and elastase that disrupt skin architecture and contribute to altered skin pigmentation (203, 204). As such, senescent cell accumulation in the skin may theoretically contribute to reduced vitamin D production, a possibility that needs to be tested. The ability of the kidneys to complete the synthesis of metabolically active vitamin D is also reduced with age (205-208). Vitamin D deficiency is widespread worldwide (209) and has been connected to obesity (210) and osteoporosis (211), among other endocrine disorders. Interestingly, vitamin D has potent geroprotective effects, with higher vitamin D levels correlating with longer telomeres in humans (212). Vitamin D has been shown to reduce senescent cell burden by inhibiting the p16 and p53 pathways (213).

Thymus

The thymus is effectively an endocrine organ as well as a vital component of the immune system since it produces hormones such as thymulin, thymosin, and thymopoietin, and the thymus is the first organ in the body that exhibits age-associated involution. Studies have shown that cellular senescence occurs in thymic epithelial cells due to high oxidative stress, especially during advanced stages of human thymic involution (214). The senescence indicators, p16^{Ink4a}, p53, p21, enhanced SA-βgal activity, and γ-H2AX were detected by immunohistochemistry in the aged human and mouse thymus (214-217). Age-related thymic involution contributes to immunosenescence and inflammaging declines due to the capacity to establish central tolerance, thereby causing increased self-reactive T cells to escape to the periphery (218). Much work remains to be done to test whether there are causal links between cellular senescence and endocrine and immune function of the thymus.

Other Organs Related to Endocrine Diseases

Vascular endothelial cells (ECs) are abundant throughout the body and produce vasoactive peptide hormones, growth factors, coagulation factors, and adhesion molecules, effectively

functioning in an endocrine-like manner. In a recent *in vitro* study, EC-conditioned media were collected from cultures treated with radiation to induce senescence in nonsenescent ECs. The senescent ECs had a more robust SASP than senescent epithelial cells or myoblasts. Senescent ECs also exhibited functional abnormalities, including decreased expression of endothelial nitric oxide synthase and increased ROS production (219). Production of prostenoids such as prostaglandin I₂ was decreased, whereas production of PAI-1, thromboxane-A₂, and endothelin-1 (was increased (220–222). Furthermore, atherosclerotic coronary arteries have increased SA- β gal (223) and *p16^{INK4a}* and *p53/p21^{CIP1/WAF1}* expression (224–228).

Senescent cells accumulate in the placenta during pregnancy (229, 230), and senescence can spread locally and systemically (231). Vascular senescence is prominent in preeclampsia, which is linked to a disrupted fetal-maternal barrier that in turn is linked to metabolic dysfunction, hypertension, recurrence of preeclampsia during subsequent pregnancies, and accelerated development of aging phenotypes in women (232–234). The effect of senescent vascular cells on the action of vasoactive and other hormones requires further study.

Therapies Targeting Senescent Cells

Cellular senescence has an important role in several conditions related to aging as well as multiple diseases across the lifespan (30, 128, 129, 204, 235). Dietary and exercise interventions can prevent the accumulation of senescent cells by reducing DNA damage, mitochondrial dysfunction, excessive ROS, and inflammation (236) or reduce deleterious properties of senescent cells, such as attenuating the SASP (237). Caloric restriction was one of the first lifespan-extending interventions to be identified. Other dietary interventions including methionine restriction (238) and ketogenic diets (239, 240) promote a favorable metabolic state that may limit the accumulation of senescent cells. Exercise is another promising strategy to delay aging phenotypes with numerous health benefits by preventing the accumulation of senescent cells (241–243). It has been shown that low-magnitude vibration, which mimics exercises, can alleviate age-related bone loss by inhibiting senescence of osteogenic cells in aged rats (244). However, whether these interventions have direct effects on senescent cells requires further study, as these interventions are multifaceted. In addition, since many reports showed that exercise activates intrinsic immune cells (245–247), these improvements on aging phenotypes may be due to the indirect removal of senescent cells by intrinsic immune cell activation (248).

Besides these nonpharmacological approaches, currently 4 approaches for targeting senescent cells are being investigated in preclinical studies (Fig. 3): (1) inhibition of senescent cell formation; (2) suppression of the SASP (senomorphics); (3) elimination of persisting senescent cells (senolytics); and (4) reprogramming of senescent cells.

Inhibition of Cellular Senescence

Targeting the cell cycle inhibitors that enforce cellular senescence directly, such as p53, which is upstream of *p21^{CIP1/WAF1}*, may prevent cells from becoming senescent. Using p53-conditional KO mice, inhibiting cellular senescence in an adipose and vascular EC-specific manner attenuated

obesity and alleviated glucose intolerance (105, 224, 225, 249). However, systemically reducing *p16^{INK4a}* or *p53-p21^{Cip1/Waf1}* expression is likely to increase cancer risk since *p53*, *p21^{Cip1/Waf1}*, and *p16^{INK4a}* global KO mice have a high prevalence of cancer and hence a shorter lifespan than control mice (250–253). Local or intermittent administration of agents that inhibit senescent cell formation might be an alternative, but even a few DNA-damaged cells with revived proliferative potential might still be sufficient to cause cancer. Hence, interfering with the capacity of cells to become senescent, as opposed to removing already senescent cells (which can include senescent cells harboring cancerous mutations), may not become a viable strategy because of the role of induction of senescence in preventing replication of cancerous cells.

Suppression of the Senescence-associated Secretory Phenotype (Senomorphics)

As some SASP factors have a role in chronic inflammation and the progression of multiple disorders, SASP inhibitors, also called senomorphic agents, can break this link between proinflammatory, proapoptotic senescent cells and disease without directly eliminating senescent cells. Several senomorphics target the transcription factor nuclear factor (NF)- κ B, Janus kinase (JAK), or the JAK signal transducer and activator of transcription (STAT) signaling pathways. Other senomorphics target rapamycin complex 1 (mammalian target of rapamycin complex 1; mTORC1), mitochondrial complex 1- or 4-related (eg, metformin), or p38 mitogen-activated protein kinase (MAPK) family members. Further possibilities for inhibiting the SASP include modulating NAD⁺/NADH metabolism, inhibiting heat shock protein 90, or neutralizing SASP factors or their receptors (254). A recently described *p21^{Cip1/Waf1}-Cre* mouse model, which has a *p21^{Cip1/Waf1}* promoter driving inducible *Cre*, enables direct targeting of *p21^{Cip1/Waf1}*-highly expressing (*p21^{high}*) senescent cells (255). Using this model, it was demonstrated that inactivating the NF- κ B pathway in *p21^{high}* cells attenuated insulin resistance in obese mice, as did the removal of *p21^{high}* cells, suggesting that SASP factors, over and above senescent cells themselves, may be a central contributor to insulin resistance (256).

Rapamycin is approved as an immunosuppressant at high doses. It or related mTORC1 inhibitors used at much lower, less immune-system-suppressing doses, appear to be senomorphic and decrease frailty (257), heart failure (258), cancer formation (259), cognitive impairment (260), immune dysfunction (261), and age-related adipose tissue loss. Rapamycin also appears to increase the maximum lifespan in mice (at least in animals raised under ideal, constant, non-stressed, pathogen-free conditions that may not reflect the real world) (262). Ruxolitinib is an inhibitor of the JAK1/JAK2-STAT3 pathway, which has been implicated in cellular senescence (263). Ruxolitinib is in clinical use for various disorders (eg, polycythemia rubra vera, myelofibrosis, and graft-versus-host disease). Ruxolitinib inhibits production of some SASP factors *in vitro* and *in vivo* in aged mice (264). In rodents, it alleviates age-related adipose tissue dysfunction, decreases insulin resistance, reduces age-related osteoporosis and frailty, alleviates critical illness myopathy, and decreases progenitor cell dysfunction (35, 265). In older myeloproliferative syndrome patients, ruxolitinib partially attenuates frailty and increases appetite, body weight, and skeletal

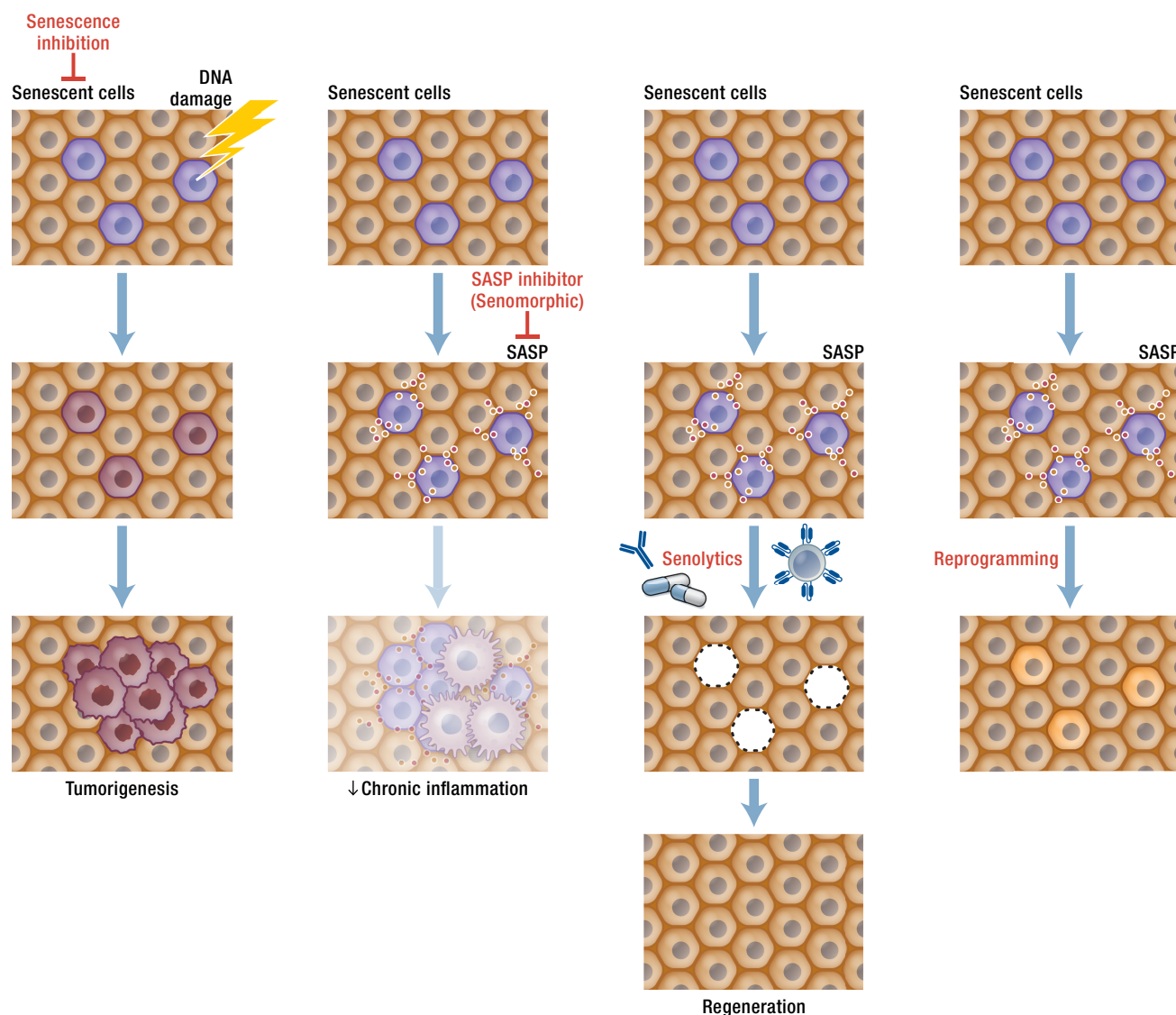


Figure 3. Scheme of senotherapies. Four strategies for attenuating detrimental effects of senescent cells. The first, targeting the senescence program itself, may lead to increased tumor formation due to apoptosis-resistant, damaged, cancerous mutation–harboring cells continuing to proliferate, and as such is not being as widely investigated currently as other approaches. The second strategy is to modify the characteristics of senescent cells using senomorphic agents that decrease production of tissue-damaging senescence-associated secretory phenotype (SASP) factors. This can be achieved using already available agents such as metformin, rapamycin, or ruxolitinib, but as the senescent cells remain in tissues, more continuous administration of senomorphic drugs may be required compared to senolytics. The third option is to target the apoptosis resistance mechanisms operative in the 30% to 70% of senescent cells that are tissue-damaging with senolytic drugs, leading to their removal. Due to the time it takes for new senescent cells to form and acquire a proapoptotic SASP, senolytics are effective even if administered intermittently using a “hit-and-run” approach. The fourth option is to alter the epigenetic programming of senescent cells by induction of Yamanaka Factors (OSKM), reverting them into a nonsenescent, replicating state. This approach, if perfected, could be promising, but since many senescent cells can harbor or develop oncogenic mutations, it could lead to cancers, including teratocarcinomas.

muscle strength, although it does not directly affect the hematological disorder itself. This indicates that ruxolitinib may alleviate frailty and geriatric phenotypes through mechanisms such as SASP inhibition independently of its effects on hematological function (266).

Metformin, an inexpensive drug that has been used to treat diabetes for more than 60 years, reduces the release of multiple proinflammatory SASP factors by senescent cells, with NF- κ B inhibition playing a key role (267). Metformin has been shown to delay, prevent, or alleviate multiple age-related disorders, including cardiovascular diseases (268), cognitive dysfunction (269), and diabetes in animals and humans (270). A retrospective analysis of patients with diabetes who

received metformin suggested there could be an increase in lifespan compared to individuals without diabetes (271). The proposed TAME (Targeting Aging with Metformin) clinical trial will test if metformin delays the appearance of a second age-related disease in patients who already have a single age-related condition (272, 273).

Disentangling the effects due to SASP modulation by these agents from other “off-target” effects that could lead to side effects is difficult. Examples of potential adverse off-target effects of senomorphics may include suppression of cytokine secretion by nonsenescent immune cells (eg, in the case of rapamycin), potentially hindering physiologically necessary inflammation in the face of an infection, or inhibition of

anabolism, potentially contributing to reduced myogenesis under some conditions. In addition, since the senescent cells themselves remain in the body, continuous administration of SASP inhibitors may be necessary, potentially leading to more side effects than senolytics, which appear to be effective even if administered intermittently. However, it should be noted that some senomorphics may be effective intermittently (274), perhaps because they decrease the SASP-induced spread of senescence (discussed next).

Elimination of Senescent Cells (Senolytics)

The idea of selectively targeting and removing senescent cells was suggested by a study that indicated interventions such as caloric restriction that enhance health span or lifespan are associated with decreased senescent cell burden in mice (173). Based on that earlier study and after work to develop senolytic agents that selectively eliminate senescent cells had already begun, this speculation was reinforced by the observation that removing cells highly expressing $p16^{Ink4a}$, many or perhaps most of which are senescent cells, enhances function in transgenic mice with an accelerated aging-like state due to transgenic expression of a gene predisposing to DNA mutations (275). In these *INK-ATTAC* mice, administration of a drug that has little or no effect on cells lacking the *ATTAC* transgene (first used in 2005 in *FAT-ATTAC* mice to remove fat cells) (276), highly expressing $p16^{Ink4a}$ cells undergo apoptosis. This observation in progeroid mice suggested that cellular senescence might be implicated in generating age-related phenotypes in naturally aging mice and that removal of senescent cells might prevent or delay cellular senescence-related dysfunction (277, 278). This was first demonstrated across multiple age-related disorders in naturally aged mice treated with senolytic agents (discussed next) that clear senescent cells (75) and confirmed in naturally aged *INK-ATTAC* mice, in which targeting highly expressing $p16^{Ink4a}$ cells alleviated senescence-related metabolic dysfunction (35). Fulfilling Koch's postulates of causality, it was shown that transplanting small numbers of senescent cells into middle-aged mice is sufficient to cause frailty, physical dysfunction, and the premature onset of most or all of the diseases that older nontransplanted mice normally die from (231). It was further shown that transplanting senescent cells causes the recipient's own cells to undergo senescence, even at a distance. Hence, senescence can spread not only in a paracrine but also distantly in an "endocrine" fashion. Additionally, intermittently eliminating senescent cells alleviated physical dysfunction after senescent cell transplantation, all pointing toward senescent cells as a causal contributor to aging phenotypes and diseases (231). Possible benefits of selectively eliminating senescent cells were subsequently indicated in mouse models using other markers of aging such as $p21^{Cip1/Waf1}$ (255, 256) and $p19$ (279). These findings support a potential role for targeting senescent cells as a therapeutic strategy to delay, prevent, alleviate, or treat multiple age-related pathologies, including endocrine and metabolic diseases and disorders (68, 129).

As senescent cells are more apoptosis-resistant than non-senescent cells (280), it was theorized that they may have increased expression of prosurvival signals, defending them against their own proapoptotic SASP factors (75). To test this, proteomic data from different types of human senescent cells were analyzed and compared to nonsenescent cells. Using bioinformatics approaches, these data were further analyzed

with the goal of identifying specific senescent cell antiapoptotic pathways (SCAPs). It was discovered that different SCAPs are upregulated depending on the senescent cell type. Next, key identified SCAP nodes were inhibited using small interfering RNAs (siRNAs). This induced apoptosis in the 30% to 70% of senescent cells that are proapoptotic and tissue-damaging (75). These SCAP components included ephrins (EFNB1 or 3)/SRC kinases, the phosphatidylinositol-4, 5-bisphosphate 3-kinase delta catalytic subunit (PI3KCD), the cyclin-dependent kinase inhibitor 1A (CDKN1A; $p21^{Cip1/Waf1}$), BCL-xL, mitochondrial pathways, and plasminogen-activated inhibitor-2 (PAI-2) (75, 281-284). Next, bioinformatics approaches were used to select small molecules that could target the identified SCAPs. Dasatinib (D), a drug approved by the Food and Drug Administration since 2006 for hematologic malignancies, is a kinase inhibitor that interferes with the SRC kinase/EFNB1/3-dependent apoptosis resistance of senescent human fat cell progenitors, leading to their selective apoptosis (285, 286). Quercetin (Q) is a natural flavonoid in apple skin, capers, and red onions that is available in the United States and Europe as a health supplement. Q is known to interfere with PI3K, and this mechanism was shown to induce death in tissue-damaging senescent human ECs (75). The combination of dasatinib and quercetin (D + Q) was theorized to have wider senolytic activity than either alone and this combination successfully reduced senescent cell burden in chronologically aged mice (75). Based on the findings regarding BCL-xL in the aforementioned siRNA studies, 8 months later 2 groups found that navitoclax (ABT-263) also exhibited senolytic activity against certain types of senescent cells and could alleviate age-related disorders and dysfunction (287, 288). To date, multiple additional senolytics have been identified using the original hypothesis-based drug discovery approach and, later, by high-throughput screening. Examples include the specific BCL-xL inhibitors A1331852 and A1155463 (289), the flavonoid fisetin (289, 290), piperlongumine (282), procyanidin C1 (291), and FOXO4-related peptide (292), among others. Several dozen small-molecule senolytics have now been identified (128). Since senescent cells can take from 1 to 6 weeks to fully develop, at least in vitro, and senescent cells do not replicate, senolytics can be administered once every couple of weeks or once a month. In mouse models, this appears to be as or more effective than administering these agents continuously (30, 128, 150).

Most current senolytics act on SCAP pathways. Second-generation senolytics are now being identified using high-throughput library screens and other approaches, including cardiac glycosides, such as digoxin, which was found to decrease the number of senescent cells after senescent cell implantation in mice (293, 294). Other recently identified senolytic targets include glutaminase 1 (295), GPNMB (glycoprotein nonmetastatic melanoma protein B) (226, 296), CD153 (297), uPAR (urokinase-type plasminogen activator receptor) (298), and intravenous zoledronic acid, a bisphosphonate that is both senolytic and senomorphic (299).

Direct In Vivo Partial Reprogramming

Another strategy targeting aging processes being tested by some research groups is cellular reprogramming using the "Yamanaka factors," Oct4, Sox2, Klf4, and c-Myc (OSKM) (300). This strategy, which uses these transcription factors

to bring senescent cells closer to a stem cell-like state, may convert the senescent cells accumulated in endocrine organs into functional cells. This reprogramming of senescent cells may be a viable treatment for several endocrine diseases, such as diabetes. In a murine model, with mice modified with an OSKM polycistronic cassette and in which OSKM expression can be induced on doxycycline treatment, OSKM induction increased pancreatic β -cell regenerative capacity and reduced metabolic dysfunction caused by streptozotocin. In older wild-type mice, reprogramming enhanced muscle regeneration after cardiotoxin-induced muscular injury (301), indicating that reprogramming may alleviate sarcopenia, much like intermittent oral senolytic D + Q administration (158). The field is still new, and it remains to be seen if it can be made safer and if creating cancers, such as malignant teratomas, by allowing potentially cancer-harboring senescent cells to reenter the cell cycle can be avoided.

Preclinical Models Targeting Endocrine Senescence

Cellular senescence has been implicated in various endocrine disorders, and preclinical data showing improvement of endocrine and related disorders with senolytics are accumulating (Table 1). Ongoing research in the senescence field may, apart from these common conditions, also possibly yield new treatment options against relatively rarer diseases, such as hypothalamic and pituitary gland disorders and aldosterone-secreting adenomas, along with several other endocrine disorders, but little is currently known about these rarer diseases as there are either no adequate preclinical models or existing models are difficult to evaluate. The evidence summarized here is for senolytics in animal models of more common conditions.

Senolytics for Diabetes Mellitus and Metabolic Syndrome

In diet-induced obese mice, senescent cell clearance has beneficial effects on adipose tissue function and systemic metabolism, with the ratio of subcutaneous to visceral fat being increased without reducing total body weight (30, 34). Senescent cell clearance also decreased lipid deposition in muscle and liver (34, 42). These changes and the extent of senescent cell clearance correlated with enhanced insulin sensitivity (34). Senescent cell clearance in aged mice mitigated age-related subcutaneous fat tissue atrophy by enhancing adipogenesis (35). In addition to these metabolic benefits, senolytics also prevented or alleviated some complications of diabetes in obese mice, including diastolic cardiac dysfunction, hepatic steatosis, microalbuminuria, and obesity-induced anxiety (34, 35, 39, 42). Clinical trials of senolytics for diabetes, adipose dysfunction, and their complications are underway (Table 2).

Senolytics and Osteoporosis, Decreased Motor Function

Intermittent senolytic treatment in old mice reduced $p16^{Ink4a}$ expression and SADS in osteocyte-enriched bone samples (78). Both genetic and pharmacologic removal of senescent cells as well as the SASP inhibitor, ruxolitinib, suppressed cortical bone resorption, increased bone formation on endocortical surfaces, and enabled bone maintenance on trabecular

surfaces of some bones. This resulted in improved bone microarchitecture and increased bone strength (44, 150). Senolytics increased osteoblast numbers and bone formation rates while reducing bone marrow adipose tissue. This was due to alterations in progenitor commitment toward osteoblast formation and away from osteoclast and adipocyte formation following senescent cell clearance. Senolytics also alleviated frailty symptoms in these mice (75, 302), indicating far-reaching effects of senescent cell removal, even when targeting a single disease. Clinical trials of senolytics for age-related osteoporosis are underway (see Table 2).

Senolytics and the Pituitary Gland

In the pituitary tumor, adamantinomatous craniopharyngioma (ACP), cells that carry oncogenic β -catenin mutations and have increased Wnt signaling, form cell clusters that become senescent and promote SASP factor production. Evidence supporting paracrine signaling by senescent cells as a risk factor for tumorigenesis across different tumors and cancer models is accumulating (303). In ex vivo cultures of pretumoral pituitary glands, the senolytic compounds navitoclax and ABT-737 tended to reduce the average size of the clusters (304) and the cardiac glycosides ouabain and digoxin, which also are senolytic compounds, killed oncogenic β -catenin-expressing cells and reduced the levels of senescence markers and SASP factors (293). These data suggest that senolytics may synergize with established anticancer drugs by eliminating OIS cells.

Senolytics and Fertility

In a small study of endometrial tissue from patients with endometriosis, D alone, Q alone, and especially D + Q reduced senescence markers and enhanced decidualization marker expression, suggesting further studies for infertility are warranted (175).

Clinical Studies of Senotherapies for Endocrine Disorders

Clinical Trials of Senolytics

Based on promising results in preclinical experiments in cultured human cells, human tissue explants, and animals, clinical studies are already underway (128). The first senolytic clinical trial published was an open-label pilot study in patients with idiopathic pulmonary fibrosis, conducted because idiopathic pulmonary fibrosis is a progressive and fatal disease with no highly effective treatment. Fourteen patients were treated with intermittent oral D + Q (100 mg/day of D and 1250 mg/day of Q) for a total of 9 doses: 3 days each week for 3 weeks in a brief, non-placebo-controlled, open-label study of safety and tolerability (305). The results suggested that senolytic treatment might alleviate physical dysfunction (gait speed, gait distance, chair stands, and short physical performance battery) in patients with idiopathic pulmonary fibrosis. However, the study was not placebo-controlled and phase 2 studies are needed. Post hoc analysis of another study involving 20 patients with idiopathic pulmonary fibrosis revealed higher levels of the geroprotective factor α -Klotho in urine after oral D + Q administration in each of the 20 participants (86).

Another pilot study involving patients with diabetic kidney disease used, for the first time, a composite score developed for

Table 1. Results from preclinical studies of senolytics in endocrine diseases

Preclinical models of therapies targeting senescent cells in endocrine organs or endocrine-related diseases			
Endocrine organ or endocrine-related disease	Model name	Treatment	Result
Pancreas	<i>INK-ATTAC</i>	Navitoclax	Improved β -cell function
Adipose tissue	<i>INK-ATTAC</i> <i>p16-3MR</i>	D + Q	Improved metabolic and adipose tissue function, reduced inflammation, improved adipogenesis
Bone	<i>INK-ATTAC</i>	D + Q	Higher bone mass and strength and better bone microarchitecture
Frailty	<i>INK-ATTAC</i>	D + Q	Improved and delayed age-associated physical dysfunction Improved muscle growth after resistance training in old age
Pituitary gland	<i>Hesx1^{Cre/+}</i> ; <i>Ctnnb1^{lox(ex3)/+}</i>	Navitoclax, ABT-737	Reduced adamantinomatous craniopharyngioma explant size after senolytic treatment
Testis	Aged	FOXO4-DRI	Improved testicular microenvironment, alleviated age-related testosterone secretion insufficiency
Ovary	Cisplatin treated	D + Q	Significant improvement of ovarian function

Abbreviation: D + Q, dasatinib and quercetin.

assaying senescent cell burden in humans. Nine individuals with diabetic kidney disease treated with a 3-day course of oral D + Q (100 mg/day of D and 1000 mg/day of Q) had decreased senescent cell burden in fat tissue compared to before the administration of senolytics (95). The fat tissue was biopsied 11 days after the last dose (95). There was also a decrease in circulating SASP factors 11 days after the last dose of D + Q compared to before D + Q was administered. Of note, D has a 3-hour and Q an 11-hour elimination half-life, so the agents were no longer present at the time of the second adipose biopsy and blood collection. This study indicated that the D + Q senolytic combination is effective in reducing senescent cell accumulation and associated inflammation in humans, suggesting that an intermittent, hit-and-run strategy may be a viable approach (95).

After these promising results, more than 30 clinical trials of senolytic therapies for a variety of diseases are planned, ongoing, or completed, including phase 2 randomized, double-blinded, placebo-controlled trials (128). Other senolytics such as fisetin as well as D + Q will be used in upcoming or ongoing phase 2 trials. If positive, these trials will have to be followed by larger clinical trials examining the effects of senolytic drugs on senescence-related disorders and diseases.

Potential Clinical Use of Senotherapies for Endocrine Diseases

Although results from preclinical data appear promising with respect to certain endocrine and related diseases, there are limited human clinical study data at this point. This is because the senotherapeutic field is new and the clinical trials conducted so far have been limited in scale since they are pilot studies. Examples of clinical trials currently planned or underway for endocrine disorders, metabolic disorders, and related diseases are in Table 2.

Following the positive results from the pilot study, fisetin as well as D + Q will be used in phase 2 trials for diabetic kidney disease. One study will examine the effect of 20 mg/kg/day of fisetin on adipose tissue-derived mesenchymal stem/stromal cell function, kidney function, systemic inflammation, and physical function in individuals with advanced chronic kidney

disease. Another study will evaluate whether targeting systemic senescent cell burden by 100 mg D + 1000 mg Q daily helps reduce markers of insulin resistance, inflammation, bone resorption, and physical dysfunction in older women with gait disturbance.

Besides diabetic kidney disease, osteoporosis is a promising target for senolytic interventions based on preclinical findings, and a clinical trial is now underway to examine age-related osteoporosis. In this trial, markers of senescent cell burden alongside bone formation and resorption markers in older women are being measured. D + Q (D 100 mg/day and Q 1000 mg/day) or fisetin (20 mg/kg/day) for 3 consecutive days is being administered to older women. Type I collagen is being assayed as a serum bone metabolism marker after 20 weeks. The Alleviation by Fisetin of Frailty, Inflammation, and Related Measures (AFFIRM) study is currently being conducted to explore effects of senolytics on musculoskeletal aging and frailty. Fisetin (20 mg/kg/day) for 3 consecutive days is being administered, and serum inflammation markers, bone resorption, insulin resistance, and gait speed are being assayed as outcomes.

In the planned “Safety and effectiveness of Quercetin & Dasatinib on epigenetic aging” study, 500 mg Q and 50 mg D for 3 days in a row per month, for a total duration of 6 months, will be administered to healthy individuals and epigenetic age will be assessed using blood samples.

NAFLD, while not a primary endocrine disorder, is a complication of obesity and insulin resistance for which effective therapeutic agents are lacking. Senolytics may be effective, as suggested by analyses in murine models that indicated decreased hepatic steatosis after senolytic treatment (25). Neither NAFLD nor the related NASH have effective treatments that can reverse these disease processes. A planned double-blinded, randomized, proof-of-principle clinical study will examine the effect of D + Q on liver fibrosis in individuals with biopsy-proven NAFLD (Table 2). In addition to senolytics, the TAME (Targeting Aging with Metformin) clinical trial is planned to test if metformin, a senomorphic drug, delays the appearance of a second age-related disease in patients without diabetes. Other clinical trials of senolytics and senomorphics are in Table 2.

Table 2. Clinical studies of senotherapies for endocrine-related diseases

Clinical studies with therapies targeting senescent cells on endocrine or endocrine-related diseases					
Targeted endocrine or endocrine-related disease	Study title	Senolytic Study	Design	Identifier	Status
Adipose tissue dysfunction	Senescence in chronic kidney diseases	D + Q	Phase 2, randomized, open-label	NCT02848131	Current, preliminary report published
Diabetic kidney disease	Inflammation and stem cells in diabetic and chronic kidney disease	Fisetin	Phase 2, randomized, double-blind, placebo-controlled	NCT03325322	Current
Insulin resistance	Alleviation by Fisetin of Frailty, Inflammation, and Related Measures in Older Women (AFFIRM)	Fisetin	Phase 2, randomized, double-blind, placebo-controlled	NCT03430037 NCT03675724	Current
Diabetic ophthalmopathy	Safety and tolerability study of UBX1325 in patients with diabetic macular edema or neovascular age-related macular degeneration	UBX1325	Phase 1, open-label Phase 2, randomized, double-blind, sham-controlled	NCT04537884 NCT04857996	Current
Osteoporosis/Frailty	Targeting cellular senescence with senolytics to improve skeletal health in older humans and Alleviation by Fisetin of Frailty, Inflammation, and Related Measures in Older Women (AFFIRM)	D + Q; Fisetin	Phase 2, randomized, open-label, Phase 2, randomized, double-blind, placebo-controlled	NCT04313634 NCT03430037 NCT03675724	Current
Epigenetic aging of all tissues	Safety and Effectiveness of Quercetin & Dasatinib on Epigenetic Aging	D + Q	Phase 1, open-label, prospective nonrandomized	NCT04946383	Active, not recruiting
NAFLD	Dasatinib and Quercetin to Treat Fibrotic Non-alcoholic Fatty Liver Disease	D + Q	Phase 1, double-blind, randomized, controlled, proof-of-principle	NCT05506488	Not yet recruiting
Platinum-resistant or refractory ovarian cancer	A Study of ABT-263 as Single Agent in Women With Platinum Resistant/Refractory Recurrent Ovarian Cancer (MONAVI-1)	Navitoclax	Phase 2, interventional single group assessment	NCT02591095	Published
Prostate cancer	Navitoclax and in Treating Patients With Progressive Metastatic Castrate Refractory Prostate Cancer	Navitoclax	Phase 2 open-label parallel assignment	NCT01828476	Terminated
Aging (diabetes, heart disease, Alzheimer, dementia, cancer)	Targeting Aging with Metformin (TAME) Trial	Metformin	Phase 3, randomized, placebo-controlled		Not yet recruiting

Abbreviation: D + Q, dasatinib and quercetin.

Senolytics may also prove to be effective for cancer-associated conditions, potentially including endocrine cancers (199, 200). As mentioned earlier, in some malignancies such as ACP, it is theorized that tumor cells can induce senescence in surrounding cells, which contributes to the malignant effects of the tumor due to SASP factor release (306). However, SASP factors can also contribute to tumor suppression by causing inflammation and recruiting immune cells that remove these damaged or oncogene-expressing cells (307). Senolytic drugs work very similarly to established anticancer treatments, and in some cases such as dasatinib and navitoclax, are repurposed anticancer drugs. There are 2 planned clinical studies using navitoclax as an adjunctive anticancer drug for cisplatin resistant/refractory ovarian cancer and metastatic castration-refractory prostate cancer. Navitoclax monotherapy has been shown to have poor activity against recurrent epithelial ovarian cancer, but there have been no unacceptable side effects. Another trial for prostate cancer was terminated early (NCT01828476). Although further research is needed on this topic and these trials were not conducted

with senolytic effects in mind, senolytic therapies may be an effective adjunctive cancer treatment after toxicity and long-term side effects have been examined.

Future Directions for Interventions Targeting Senescent Cells

The senolytic treatments in many of the aforementioned clinical studies have to date had only mild side effects and off-target effects. To balance risk with potential benefits, so far clinical trials have focused on serious disorders and diseases for which few treatment options exist, rather than preventive studies in healthy individuals. In addition, there are 2 major issues to be solved before getting to the point where the clinical use of senotherapies can be advocated. First, even though some of the early clinical trials have suggested short-term safety and target engagement for senolytics, the field is still new. Larger, randomized clinical trials are needed for detecting any longer-term adverse effects and determining if these agents are effective. In our opinion, the only context for

administering these agents is in carefully controlled clinical trials with data monitoring and safety oversight. Senolytics and other senotherapies should not be prescribed or used over the counter. There is a possibility that many of the early studies will fail. These trials should be viewed as a first step and a basis for continuous improvement of trial design so that as better treatments are developed, these trials can be used as a template. Several of the agents being administered in these early studies are approved for other uses in humans or are natural products with robust safety data. In our view, this needs to be the initial approach rather than beginning with new drugs rarely used in humans before. With these cautions, setbacks of the type that slowed clinical translation of gene therapies due to serious side effects in early trials might be avoided.

The second issue is gerodiagnostic tests. To facilitate the development of interventions, further work in refining gerodiagnostic tests measuring senescent cell burden and the extent of other fundamental aging processes is required. Mere “aging clocks” will be less informative than signatures or composite scores of analytes that change in response to interventions, predict and track clinical changes, indicate which intervention to use, are reliable, reproducible, scalable, and inexpensive, and can be assayed in urine, saliva, blood, buccal swabs, or other easily obtained samples. There is not any single, universal, or fully specific biomarker for identifying senescent cells reliably in cell culture or tissue samples because the phenotype of senescent cells varies considerably among cell types and tissues. New approaches are being explored that take the initial cell type into account to subclassify senescent cells and focus on distinct senescent cell types. The initial trigger for senescence may determine the phenotype of senescent cells, an example being adipocyte senescence induced by high-fat diets, in which upregulation of p21^{CIP1/WAF1} but not so much of p16^{INK4a} occurs (255, 256). To characterize senescence fully and develop more sensitive and specific markers of different senescent cell subtypes, transcriptomic and proteomic studies down to the single-cell level across relevant cell and tissue types will be of importance, facilitating delineation of cell surface molecules that will allow detection and isolation of senescent cells from blood or tissue samples. Another solution to this problem is to combine sets of SASP factors and other senescence-associated analytes into a composite score to identify and quantify senescent cells. Several such panels have been developed, the first being one that successfully tracked decreases in senescent cell burden in a clinical trial of senolytics (95). Frequent components of these panels are p21^{CIP1/WAF1} or p16^{INK4a} levels, along with inflammatory SASP factors such as TNF α or IL-1 α . Another such composite score is the “SenMayo” panel, which appears to identify senescent cells across different tissues and species (96). This panel can be used at the tissue and single-cell level and identifies key signaling pathways. Panels such as SenMayo may facilitate monitoring senescent cell burden with aging and in diseases and for analyzing the effect of therapies targeting cellular senescence during clinical trials.

Conclusions

In recent years, research targeting fundamental aging processes has made remarkable progress, and senescent cell-targeting therapies including SASP inhibitors, senolytics, and reprogramming are promising. With accumulating evidence

from preclinical studies, some of these therapies have progressed to the point of early clinical trials. Diabetes, metabolic syndrome, macular degeneration, and osteoporosis are among endocrine and endocrine-related disorders for which clinical senolytic trials are planned or ongoing.

Senotherapies are a new treatment strategy and have the potential to significantly change existing treatments. Targeting senescent cells is an intervention against a root-cause contributor to the diseases of aging and may improve age-related conditions in a comprehensive manner, rather than one disease at a time. On the other hand, this field is new, and there are many issues such as long-term outcomes, side effects, and the establishment of gerodiagnostics, so we must proceed carefully. Adoption by the Food and Drug Administration or the World Health Organization International Classification of Diseases code for impaired functional capacity would help in obtaining and tracking data at an epidemiological level and facilitating regulatory oversight. If interventions or combinations or interventions (eg, lifestyle plus drug) can be developed that target interlinked fundamental aging processes additively or synergistically, their first use will likely be for serious diseases and disorders linked to these processes. Eventually, if safe and effective, senotherapies may move toward use for less-serious conditions, as agents coadministered with other interventions already developed to treat particular diseases, and perhaps, eventually, for secondary prevention and even primary prevention to enhance health span.

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Author Contributions

J.L.K. conceptualized this review. All authors contributed to the writing and approved the submitted version of the manuscript.

Disclosures

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References

1. WHO. Ageing and health. Accessed April 5, 2024. <https://www.who.int/news-room/fact-sheets/detail/ageing-and-health>
2. Niccoli T, Partridge L. Ageing as a risk factor for disease. *Curr Biol*. 2012;22(17):R741-R752.

3. Cappola AR, Auchus RJ, El-Hajj Fuleihan G, *et al.* Hormones and aging: an endocrine society scientific statement. *J Clin Endocrinol Metab.* 2023;108(8):1835-1874.
4. Horstman AM, Dillon EL, Urban RJ, Sheffield-Moore M. The role of androgens and estrogens on healthy aging and longevity. *J Gerontol A Biol Sci Med Sci.* 2012;67(11):1140-1152.
5. Orentreich N, Brind JL, Rizer RL, Vogelmann JH. Age changes and sex differences in serum dehydroepiandrosterone sulfate concentrations throughout adulthood. *J Clin Endocrinol Metab.* 1984;59(3):551-555.
6. Baulieu EE. Dehydroepiandrosterone (DHEA): a fountain of youth? *J Clin Endocrinol Metab.* 1996;81(9):3147-3151.
7. Giustina A, Veldhuis JD. Pathophysiology of the neuroregulation of growth hormone secretion in experimental animals and the human. *Endocr Rev.* 1998;19(6):717-797.
8. Nass R, Farhy LS, Liu J, *et al.* Age-dependent decline in acyl-ghrelin concentrations and reduced association of acyl-ghrelin and growth hormone in healthy older adults. *J Clin Endocrinol Metab.* 2014;99(2):602-608.
9. Rudman D, Kutner MH, Rogers CM, Lubin MF, Fleming GA, Bain RP. Impaired growth hormone secretion in the adult population: relation to age and adiposity. *J Clin Invest.* 1981;67(5):1361-1369.
10. Ho KY, Evans WS, Blizzard RM, *et al.* Effects of sex and age on the 24-hour profile of growth hormone secretion in man: importance of endogenous estradiol concentrations. *J Clin Endocrinol Metab.* 1987;64(1):51-58.
11. Hegstad R, Brown RD, Jiang NS, *et al.* Aging and aldosterone. *Am J Med.* 1983;74(3):442-448.
12. Karasek M. Melatonin, human aging, and age-related diseases. *Exp Gerontol.* 2004;39(11-12):1723-1729.
13. Martín Giménez VM, de Las Heras N, Lahera V, Tresguerres JAF, Reiter RJ, Manucha W. Melatonin as an anti-aging therapy for age-related cardiovascular and neurodegenerative diseases. *Front Aging Neurosci.* 2022;14:888292.
14. Mishra SR, Chung HF, Waller M, Mishra GD. Duration of estrogen exposure during reproductive years, age at menarche and age at menopause, and risk of cardiovascular disease events, all-cause and cardiovascular mortality: a systematic review and meta-analysis. *BJOG.* 2021;128(5):809-821.
15. Lobo RA, Davis SR, De Villiers TJ, *et al.* Prevention of diseases after menopause. *Climacteric.* 2014;17(5):540-556.
16. Deuschle M, Gotthardt U, Schweiger U, *et al.* With aging in humans the activity of the hypothalamus-pituitary-adrenal system increases and its diurnal amplitude flattens. *Life Sci.* 1997;61(22):2239-2246.
17. Veldhuis JD, Sharma A, Roelfsema F. Age-dependent and gender-dependent regulation of hypothalamic-adrenocorticotrophic-adrenal axis. *Endocrinol Metab Clin North Am.* 2013;42(2):201-225.
18. Purnell JQ, Brandon DD, Isabelle LM, Loriaux DL, Samuels MH. Association of 24-hour cortisol production rates, cortisol-binding globulin, and plasma-free cortisol levels with body composition, leptin levels, and aging in adult men and women. *J Clin Endocrinol Metab.* 2004;89(1):281-287.
19. Waring AC, Arnold AM, Newman AB, Buzková P, Hirsch C, Cappola AR. Longitudinal changes in thyroid function in the oldest old and survival: the cardiovascular health study all-stars study. *J Clin Endocrinol Metab.* 2012;97(11):3944-3950.
20. van den Beld AW, Kaufman JM, Zillikens MC, Lamberts SWJ, Egan JM, van der Lely AJ. The physiology of endocrine systems with ageing. *Lancet Diabetes Endocrinol.* 2018;6(8):647-658.
21. Bremner AP, Feddema P, Leedman PJ, *et al.* Age-related changes in thyroid function: a longitudinal study of a community-based cohort. *J Clin Endocrinol Metab.* 2012;97(5):1554-1562.
22. Surks MI, Hollowell JG. Age-specific distribution of serum thyrotropin and antithyroid antibodies in the US population: implications for the prevalence of subclinical hypothyroidism. *J Clin Endocrinol Metab.* 2007;92(12):4575-4582.
23. Al-Sofiani ME, Ganji SS, Kalyani RR. Body composition changes in diabetes and aging. *J Diabetes Complications.* 2019;33(6):451-459.
24. Santen RJ, Allred DC, Ardoin SP, *et al.* Postmenopausal hormone therapy: an Endocrine Society scientific statement. *J Clin Endocrinol Metab.* 2010;95(7 Suppl 1):s1-s66.
25. Basaria S, Coviello AD, Travison TG, *et al.* Adverse events associated with testosterone administration. *N Engl J Med.* 2010;363(2):109-122.
26. Liu H, Bravata DM, Olkin I, *et al.* Systematic review: the safety and efficacy of growth hormone in the healthy elderly. *Ann Intern Med.* 2007;146(2):104-115.
27. Carto CA, Gurayah AA, Arbelaez MCS, Grewal MR, Kohn T, Ramasamy R. Association between comorbidities and longitudinal changes in total testosterone among men from the Baltimore longitudinal study of aging. *J Sex Med.* 2023;20(5):605-611.
28. Nanba K, Vaidya A, Williams GH, Zheng I, Else T, Rainey WE. Age-related autonomous aldosteronism. *Circulation.* 2017;136(4):347-355.
29. Gunasekaran U, Gannon M. Type 2 diabetes and the aging pancreatic beta cell. *Aging (Albany NY).* 2011;3(6):565-575.
30. Palmer AK, Tchkonja T, Kirkland JL. Targeting cellular senescence in metabolic disease. *Mol Metab.* 2022;66:101601.
31. Wang L, Wang B, Gasek NS, *et al.* Targeting p21(Cip1) highly expressing cells in adipose tissue alleviates insulin resistance in obesity. *Cell Metab.* 2022;34(1):75-89.e8.
32. Chaib S, Tchkonja T, Kirkland JL. Obesity, senescence, and senolytics. *Handb Exp Pharmacol.* 2022;274:165-180.
33. Conley SM, Hickson LJ, Kellogg TA, *et al.* Human obesity induces dysfunction and early senescence in adipose tissue-derived mesenchymal stromal/stem cells. *Front Cell Dev Biol.* 2020;8:197.
34. Palmer AK, Xu M, Zhu Y, *et al.* Targeting senescent cells alleviates obesity-induced metabolic dysfunction. *Aging Cell.* 2019;18(3):e12950.
35. Xu M, Palmer AK, Ding H, *et al.* Targeting senescent cells enhances adipogenesis and metabolic function in old age. *Elife.* 2015;4:e12997.
36. Escande C, Nin V, Pirtskhalava T, *et al.* Deleted in breast cancer 1 regulates cellular senescence during obesity. *Aging Cell.* 2014;13(5):951-953.
37. Tchkonja T, Morbeck DE, Von Zglinicki T, *et al.* Fat tissue, aging, and cellular senescence. *Aging Cell.* 2010;9(5):667-684.
38. Bian X, Griffin TP, Zhu X, *et al.* Senescence marker activin A is increased in human diabetic kidney disease: association with kidney function and potential implications for therapy. *BMJ Open Diabetes Res Care.* 2019;7(1):e000720.
39. Kim SR, Jiang K, Ogrodnik M, *et al.* Increased renal cellular senescence in murine high-fat diet: effect of the senolytic drug quercetin. *Transl Res.* 2019;213:112-123.
40. Meijnikman AS, van Olden CC, Aydin Ö, *et al.* Hyperinsulinemia is highly associated with markers of hepatocytic senescence in two independent cohorts. *Diabetes.* 2022;71(9):1929-1936.
41. Baboota RK, Rawshani A, Bonnet L, *et al.* BMP4 and Gremlin 1 regulate hepatic cell senescence during clinical progression of NAFLD/NASH. *Nat Metab.* 2022;4(8):1007-1021.
42. Ogrodnik M, Miwa S, Tchkonja T, *et al.* Cellular senescence drives age-dependent hepatic steatosis. *Nat Commun.* 2017;8(1):15691.
43. Farr JN, Rowsey JL, Eckhardt BA, *et al.* Independent roles of estrogen deficiency and cellular senescence in the pathogenesis of osteoporosis: evidence in young adult mice and older humans. *J Bone Miner Res.* 2019;34(8):1407-1418.
44. Khosla S, Farr JN, Kirkland JL. Inhibiting cellular senescence: a new therapeutic paradigm for age-related osteoporosis. *J Clin Endocrinol Metab.* 2018;103(4):1282-1290.
45. Chandra A, Lagnado AB, Farr JN, *et al.* Targeted reduction of senescent cell burden alleviates focal radiotherapy-related bone loss. *J Bone Miner Res.* 2020;35(6):1119-1131.

46. Chandra A, Lagnado AB, Farr JN, *et al.* Targeted clearance of p21- but not p16-positive senescent cells prevents radiation-induced osteoporosis and increased marrow adiposity. *Aging Cell*. 2022;21(5):e13602.
47. Cavalcante MB, Saccon TD, Nunes ADC, *et al.* Dasatinib plus quercetin prevents uterine age-related dysfunction and fibrosis in mice. *Aging (Albany NY)*. 2020;12(3):2711-2722.
48. de Magalhães JP. How ageing processes influence cancer. *Nat Rev Cancer*. 2013;13(5):357-365.
49. Cupit-Link MC, Kirkland JL, Ness KK, *et al.* Biology of premature ageing in survivors of cancer. *ESMO Open*. 2017;2(5):e000250.
50. North BJ, Sinclair DA. The intersection between aging and cardiovascular disease. *Circ Res*. 2012;110(8):1097-1108.
51. Lewis-McDougall FC, Ruchaya PJ, Domenjo-Vila E, *et al.* Aged-senescent cells contribute to impaired heart regeneration. *Aging Cell*. 2019;18(3):e12931.
52. Yu S, Kim SR, Jiang K, *et al.* Quercetin reverses cardiac systolic dysfunction in mice fed with a high-fat diet: role of angiogenesis. *Oxid Med Cell Longev*. 2021;2021:8875729.
53. Ogrodnik M, Zhu Y, Langhi LGP, *et al.* Obesity-induced cellular senescence drives anxiety and impairs neurogenesis. *Cell Metab*. 2019;29(5):1061-1077.e8.
54. Hayflick L, Moorhead PS. The serial cultivation of human diploid cell strains. *Exp Cell Res*. 1961;25(3):585-621.
55. Campisi J, d'Adda di Fagnana F. Cellular senescence: when bad things happen to good cells. *Nat Rev Mol Cell Biol*. 2007;8(9):729-740.
56. Young AR, Narita M. SASP reflects senescence. *EMBO Rep*. 2009;10(3):228-230.
57. Coppé JP, Patil CK, Rodier F, *et al.* Senescence-associated secretory phenotypes reveal cell-nonautonomous functions of oncogenic RAS and the p53 tumor suppressor. *PLoS Biol*. 2008;6(12):2853-2868.
58. Tchkonja T, Zhu Y, van Deursen J, Campisi J, Kirkland JL. Cellular senescence and the senescent secretory phenotype: therapeutic opportunities. *J Clin Invest*. 2013;123(3):966-972.
59. Basisty N, Kale A, Jeon OH, *et al.* A proteomic atlas of senescence-associated secretomes for aging biomarker development. *PLoS Biol*. 2020;18(1):e3000599.
60. Hernandez-Segura A, de Jong TV, Melov S, Guryev V, Campisi J, Demaria M. Unmasking transcriptional heterogeneity in senescent cells. *Curr Biol*. 2017;27(17):2652-2660.e4.
61. Wiley CD, Flynn JM, Morrissey C, *et al.* Analysis of individual cells identifies cell-to-cell variability following induction of cellular senescence. *Aging Cell*. 2017;16(5):1043-1050.
62. Jochems F, Thijssen B, De Conti G, *et al.* The cancer SENESCopedia: a delineation of cancer cell senescence. *Cell Rep*. 2021;36(4):109441.
63. Tripathi U, Misra A, Tchkonja T, Kirkland JL. Impact of senescent cell subtypes on tissue dysfunction and repair: importance and research questions. *Mech Ageing Dev*. 2021;198:111548.
64. Hudgins AD, Tazearslan C, Tare A, Zhu Y, Huffman D, Suh Y. Age- and tissue-specific expression of senescence biomarkers in mice. *Front Genet*. 2018;9:59.
65. Kennedy BK, Berger SL, Brunet A, *et al.* Geroscience: linking aging to chronic disease. *Cell*. 2014;159(4):709-713.
66. Inouye SK, Studenski S, Tinetti ME, Kuchel GA. Geriatric syndromes: clinical, research, and policy implications of a core geriatric concept. *J Am Geriatr Soc*. 2007;55(5):780-791.
67. Hadley EC, Kuchel GA, Newman AB. Report: NIA workshop on measures of physiologic resiliencies in human aging. *J Gerontol A Biol Sci Med Sci*. 2017;72(7):980-990.
68. Raffaele M, Vinciguerra M. The costs and benefits of senotherapeutics for human health. *Lancet Healthy Longev*. 2022;3(1):e67-e77.
69. López-Otín C, Blasco MA, Partridge L, Serrano M, Kroemer G. The hallmarks of aging. *Cell*. 2013;153(6):1194-1217.
70. Sinclair AJ, Conroy SP, Bayer AJ. Impact of diabetes on physical function in older people. *Diabetes Care*. 2008;31(2):233-235.
71. Corriere M, Rooparinesingh N, Kalyani RR. Epidemiology of diabetes and diabetes complications in the elderly: an emerging public health burden. *Curr Diab Rep*. 2013;13(6):805-813.
72. Kalyani RR, Metter EJ, Ramachandran R, Chia CW, Saudek CD, Ferrucci L. Glucose and insulin measurements from the oral glucose tolerance test and relationship to muscle mass. *J Gerontol A Biol Sci Med Sci*. 2012;67(1):74-81.
73. Dimri GP, Lee X, Basile G, *et al.* A biomarker that identifies senescent human cells in culture and in aging skin in vivo. *Proc Natl Acad Sci U S A*. 1995;92(20):9363-9367.
74. González-Gualda E, Baker AG, Fruk L, Muñoz-Espín D. A guide to assessing cellular senescence in vitro and in vivo. *FEBS J*. 2021;288(1):56-80.
75. Zhu Y, Tchkonja T, Pirtskhalava T, *et al.* The Achilles' heel of senescent cells: from transcriptome to senolytic drugs. *Aging Cell*. 2015;14(4):644-658.
76. Swanson EC, Manning B, Zhang H, Lawrence JB. Higher-order unfolding of satellite heterochromatin is a consistent and early event in cell senescence. *J Cell Biol*. 2013;203(6):929-942.
77. Narita M, Nunez S, Heard E, *et al.* Rb-mediated heterochromatin formation and silencing of E2F target genes during cellular senescence. *Cell*. 2003;113(6):703-716.
78. Farr JN, Fraser DG, Wang H, *et al.* Identification of senescent cells in the bone microenvironment. *J Bone Miner Res*. 2016;31(11):1920-1929.
79. Iske J, Seyda M, Heinbokel T, *et al.* Senolytics prevent mt-DNA-induced inflammation and promote the survival of aged organs following transplantation. *Nat Commun*. 2020;11(1):4289.
80. Papaconstantinou J. Insulin/IGF-1 and ROS signaling pathway cross-talk in aging and longevity determination. *Mol Cell Endocrinol*. 2009;299(1):89-100.
81. Daneshgar N, Dai DF. ROS, Klotho and mTOR in cardiorenal aging. *Aging (Albany NY)*. 2020;12(20):19830-19831.
82. Wu C, Ma X, Zhou Y, Liu Y, Shao Y, Wang Q. Klotho restraining Egr1/TLR4/mTOR axis to reducing the expression of fibrosis and inflammatory cytokines in high glucose cultured rat mesangial cells. *Exp Clin Endocrinol Diabetes*. 2019;127(9):630-640.
83. Rakugi H, Matsukawa N, Ishikawa K, *et al.* Anti-oxidative effect of Klotho on endothelial cells through cAMP activation. *Endocrine*. 2007;31(1):82-87.
84. Xie L, Wang Y, Li Q, *et al.* The HIF-1 α /p53/miRNA-34a/Klotho axis in retinal pigment epithelial cells promotes subretinal fibrosis and exacerbates choroidal neovascularization. *J Cell Mol Med*. 2021;25(3):1700-1711.
85. Kuro OM. The Klotho proteins in health and disease. *Nat Rev Nephrol*. 2019;15(1):27-44.
86. Zhu Y, Prata L, Gerdes EOW, *et al.* Orally-active, clinically-translatable senolytics restore α -Klotho in mice and humans. *EBioMedicine*. 2022;77:103912.
87. Munk R, Panda AC, Grammatikakis I, Gorospe M, Abdelmohsen K. Senescence-associated microRNAs. *Int Rev Cell Mol Biol*. 2017;334:177-205.
88. Suh N. MicroRNA controls of cellular senescence. *BMB Rep*. 2018;51(10):493-499.
89. Tanaka Y, Takahashi A. Senescence-associated extracellular vesicle release plays a role in senescence-associated secretory phenotype (SASP) in age-associated diseases. *J Biochem*. 2021;169(2):147-153.
90. Misawa T, Hitomi K, Miyata K, *et al.* Identification of novel senescent markers in small extracellular vesicles. *Int J Mol Sci*. 2023;24(3):2421.
91. Faraonio R, Salerno P, Passaro F, *et al.* A set of miRNAs participates in the cellular senescence program in human diploid fibroblasts. *Cell Death Differ*. 2012;19(4):713-721.
92. Mori MA, Raghavan P, Thomou T, *et al.* Role of microRNA processing in adipose tissue in stress defense and longevity. *Cell Metab*. 2012;16(3):336-347.

93. Gorgoulis V, Adams PD, Alimonti A, *et al.* Cellular senescence: defining a path forward. *Cell*. 2019;179(4):813-827.
94. Biran A, Zada L, Abou Karam P, *et al.* Quantitative identification of senescent cells in aging and disease. *Aging Cell*. 2017;16(4):661-671.
95. Hickson LJ, Langhi Prata LGP, Bobart SA, *et al.* Senolytics decrease senescent cells in humans: preliminary report from a clinical trial of Dasatinib plus Quercetin in individuals with diabetic kidney disease. *EBioMedicine*. 2019;47:446-456.
96. Saul D, Kosinsky RL, Atkinson EJ, *et al.* A new gene set identifies senescent cells and predicts senescence-associated pathways across tissues. *Nat Commun*. 2022;13(1):4827.
97. Liu Y, Sanoff HK, Cho H, *et al.* Expression of p16(INK4a) in peripheral blood T-cells is a biomarker of human aging. *Aging Cell*. 2009;8(4):439-448.
98. Englund DA, Sakamoto AE, Fritsche CM, *et al.* Exercise reduces circulating biomarkers of cellular senescence in humans. *Aging Cell*. 2021;20(7):e13415.
99. CDC. National diabetes statistics report. Accessed April 5, 2024. <https://www.cdc.gov/diabetes/data/statistics-report/index.html>
100. Kalyani RR, Egan JM. Diabetes and altered glucose metabolism with aging. *Endocrinol Metab Clin North Am*. 2013;42(2):333-347.
101. Hauner H. Secretory factors from human adipose tissue and their functional role. *Proc Nutr Soc*. 2005;64(2):163-169.
102. Halberg N, Wernstedt-Asterholm I, Scherer PE. The adipocyte as an endocrine cell. *Endocrinol Metab Clin North Am*. 2008;37(3):753-768, x-xi.
103. Coelho M, Oliveira T, Fernandes R. Biochemistry of adipose tissue: an endocrine organ. *Arch Med Sci*. 2013;9(2):191-200.
104. Kirkland JL, Tchkonja T, Pirtskhalava T, Han J, Karagiannides I. Adipogenesis and aging: does aging make fat go MAD? *Exp Gerontol*. 2002;37(6):757-767.
105. Minamino T, Orimo M, Shimizu I, *et al.* A crucial role for adipose tissue p53 in the regulation of insulin resistance. *Nat Med*. 2009;15(9):1082-1087.
106. Villaret A, Galitzky J, Decaunes P, *et al.* Adipose tissue endothelial cells from obese human subjects: differences among depots in angiogenic, metabolic, and inflammatory gene expression and cellular senescence. *Diabetes*. 2010;59(11):2755-2763.
107. Dhirachakulpanich D, Lagger C, Chatsirisupachai K, de Magalhães JP, Paraoan L. Intercellular communication analysis of the human retinal pigment epithelial and choroidal cells predicts pathways associated with aging, cellular senescence and age-related macular degeneration. *Front Aging Neurosci*. 2022;14:1016293.
108. Chae JB, Park CW, Lee HM, *et al.* Accelerated aging phenotypes in the retinal pigment epithelium of Zmpste24-deficient mice. *Biochem Biophys Res Commun*. 2022;632:62-68.
109. Russell SJ, Kahn CR. Endocrine regulation of ageing. *Nat Rev Mol Cell Biol*. 2007;8(9):681-691.
110. Kimura KD, Tissenbaum HA, Liu Y, Ruvkun G. daf-2, an insulin receptor-like gene that regulates longevity and diapause in *Caenorhabditis elegans*. *Science*. 1997;277(5328):942-946.
111. Kenyon C, Chang J, Gensch E, Rudner A, Tabtiang R. A *C. elegans* mutant that lives twice as long as wild type. *Nature*. 1993;366(6454):461-464.
112. Minamino T, Komuro I. Vascular cell senescence: contribution to atherosclerosis. *Circ Res*. 2007;100(1):15-26.
113. Hwangbo DS, Gershman B, Tu MP, Palmer M, Tatar M. Drosophila dFOXO controls lifespan and regulates insulin signaling in brain and fat body. *Nature*. 2004;429(6991):562-566.
114. Giannakou ME, Goss M, Jünger MA, Hafen E, Leivers SJ, Partridge L. Long-lived *Drosophila* with overexpressed dFOXO in adult fat body. *Science*. 2004;305(5682):361.
115. Fabrizio P, Pozza F, Pletcher SD, Gendron CM, Longo VD. Regulation of longevity and stress resistance by Sch9 in yeast. *Science*. 2001;292(5515):288-290.
116. Fabrizio P, Liou LL, Moy VN, *et al.* SOD2 functions downstream of Sch9 to extend longevity in yeast. *Genetics*. 2003;163(1):35-46.
117. Clancy DJ, Gems D, Harshman LG, *et al.* Extension of life-span by loss of CHICO, a *Drosophila* insulin receptor substrate protein. *Science*. 2001;292(5514):104-106.
118. Holzenberger M, Dupont J, Ducos B, *et al.* IGF-1 receptor regulates lifespan and resistance to oxidative stress in mice. *Nature*. 2003;421(6919):182-187.
119. Blüher M, Kahn BB, Kahn CR. Extended longevity in mice lacking the insulin receptor in adipose tissue. *Science*. 2003;299(5606):572-574.
120. Taguchi A, Wartschow LM, White MF. Brain IRS2 signaling coordinates life span and nutrient homeostasis. *Science*. 2007;317(5836):369-372.
121. Aguiar-Oliveira MH, Bartke A. Growth hormone deficiency: health and longevity. *Endocr Rev*. 2019;40(2):575-601.
122. Colman RJ, Anderson RM, Johnson SC, *et al.* Caloric restriction delays disease onset and mortality in rhesus monkeys. *Science*. 2009;325(5937):201-204.
123. Ramsey JJ, Colman RJ, Binkley NC, *et al.* Dietary restriction and aging in rhesus monkeys: the university of Wisconsin study. *Exp Gerontol*. 2000;35(9-10):1131-1149.
124. Gresl TA, Colman RJ, Roecker EB, *et al.* Dietary restriction and glucose regulation in aging rhesus monkeys: a follow-up report at 8.5yr. *Am J Physiol Endocrinol Metab*. 2001;281(4):E757-E765.
125. Roth GS, Lane MA, Ingram DK, *et al.* Biomarkers of caloric restriction may predict longevity in humans. *Science*. 2002;297(5582):811.
126. Ben-Avraham D, Govindaraju DR, Budagov T, *et al.* The GH receptor exon 3 deletion is a marker of male-specific exceptional longevity associated with increased GH sensitivity and taller stature. *Sci Adv*. 2017;3(6):e1602025.
127. van der Spoel E, Jansen SW, Akintola AA, *et al.* Growth hormone secretion is diminished and tightly controlled in humans enriched for familial longevity. *Aging Cell*. 2016;15(6):1126-1131.
128. Chaib S, Tchkonja T, Kirkland JL. Cellular senescence and senolytics: the path to the clinic. *Nat Med*. 2022;28(8):1556-1568.
129. Tchkonja T, Palmer AK, Kirkland JL. New horizons: novel approaches to enhance healthspan through targeting cellular senescence and related aging mechanisms. *J Clin Endocrinol Metab*. 2021;106(3):e1481-e1487.
130. Conover CA, Bale LK. Senescence induces proteolytically-active PAPP-A secretion and association with extracellular vesicles in human pre-adipocytes. *Exp Gerontol*. 2022;172:112070.
131. Conover CA. Key questions and answers about pregnancy-associated plasma protein-A. *Trends Endocrinol Metab*. 2012;23(5):242-249.
132. Bale LK, West SA, Conover CA. Inducible knockdown of pregnancy-associated plasma protein-A gene expression in adult female mice extends life span. *Aging Cell*. 2017;16(4):895-897.
133. Conover CA, Bale LK, Marler RJ. Pregnancy-associated plasma protein-A deficiency improves survival of mice on a high fat diet. *Exp Gerontol*. 2015;70:131-134.
134. Conover CA, Harstad SL, Tchkonja T, Kirkland JL. Preferential impact of pregnancy-associated plasma protein-A deficiency on visceral fat in mice on high-fat diet. *Am J Physiol Endocrinol Metab*. 2013;305(9):E1145-E1153.
135. Bian A, Ma Y, Zhou X, *et al.* Association between sarcopenia and levels of growth hormone and insulin-like growth factor-1 in the elderly. *BMC Musculoskelet Disord*. 2020;21(1):214.
136. Ascenzi F, Barberi L, Dobrowolny G, *et al.* Effects of IGF-1 isoforms on muscle growth and sarcopenia. *Aging Cell*. 2019;18(3):e12954.
137. Chakravarthy MV, Davis BS, Booth FW. IGF-I restores satellite cell proliferative potential in immobilized old skeletal muscle. *J Appl Physiol* (1985). 2000;89(4):1365-1379.
138. Latres E, Amini AR, Amini AA, *et al.* Insulin-like growth factor-1 (IGF-1) inversely regulates atrophy-induced genes via the

- phosphatidylinositol 3-kinase/Akt/mammalian target of rapamycin (PI3K/Akt/mTOR) pathway. *J Biol Chem.* 2005;280(4):2737-2744.
139. O'Neill BT, Lee KY, Klaus K, *et al.* Insulin and IGF-1 receptors regulate FoxO-mediated signaling in muscle proteostasis. *J Clin Invest.* 2016;126(9):3433-3446.
 140. Rudman D, Feller AG, Nagraj HS, *et al.* Effects of human growth hormone in men over 60 years old. *N Engl J Med.* 1990;323(1):1-6.
 141. Beigienė A, Petruševičienė D, Barasaitė V, Kubilius R, Macijauskienė J. Frailty and different exercise interventions to improve gait speed in older adults after acute coronary syndrome. *Medicina (Kaunas).* 2021;57(12):1344.
 142. Jamali T, Raasikh T, Bustamante G, *et al.* Outcomes of exercise interventions in patients with advanced liver disease: a systematic review of randomized clinical trials. *Am J Gastroenterol.* 2022;117(10):1614-1620.
 143. Flor-Rufino C, Barrachina-Igual J, Pérez-Ros P, Pablos-Monzó A, Martínez-Arnau FM. Resistance training of peripheral muscles benefits respiratory parameters in older women with sarcopenia: randomized controlled trial. *Arch Gerontol Geriatr.* 2022;104:104799.
 144. Swales B, Ryde GC, Whittaker AC. A randomized controlled feasibility trial evaluating a resistance training intervention with frail older adults in residential care: the keeping active in residential elderly trial. *J Aging Phys Act.* 2022;30(3):364-388.
 145. Bolster DR, Jefferson LS, Kimball SR. Regulation of protein synthesis associated with skeletal muscle hypertrophy by insulin-, amino acid- and exercise-induced signalling. *Proc Nutr Soc.* 2004;63(2):351-356.
 146. Ensrud KE. Epidemiology of fracture risk with advancing age. *J Gerontol A Biol Sci Med Sci.* 2013;68(10):1236-1242.
 147. Sambrook P, Cooper C. Osteoporosis. *Lancet.* 2006;367(9527):2010-2018.
 148. MacLaughlin J, Holick MF. Aging decreases the capacity of human skin to produce vitamin D3. *J Clin Invest.* 1985;76(4):1536-1538.
 149. Franco AC, Aveleira C, Cavadas C. Skin senescence: mechanisms and impact on whole-body aging. *Trends Mol Med.* 2022;28(2):97-109.
 150. Farr JN, Xu M, Weivoda MM, *et al.* Targeting cellular senescence prevents age-related bone loss in mice. *Nat Med.* 2017;23(9):1072-1079.
 151. Khosla S, Farr JN, Tchkonja T, Kirkland JL. The role of cellular senescence in ageing and endocrine disease. *Nat Rev Endocrinol.* 2020;16(5):263-275.
 152. Yang SB, Tien AC, Boddupalli G, Xu AW, Jan YN, Jan LY. Rapamycin ameliorates age-dependent obesity associated with increased mTOR signaling in hypothalamic POMC neurons. *Neuron.* 2012;75(3):425-436.
 153. Veldhuis JD. Changes in pituitary function with ageing and implications for patient care. *Nat Rev Endocrinol.* 2013;9(4):205-215.
 154. Morimoto N, Kawakami F, Makino S, Chihara K, Hasegawa M, Ibata Y. Age-related changes in growth hormone releasing factor and somatostatin in the rat hypothalamus. *Neuroendocrinology.* 1988;47(5):459-464.
 155. Farhy LS, Veldhuis JD. Deterministic construct of amplifying actions of ghrelin on pulsatile growth hormone secretion. *Am J Physiol Regul Integr Comp Physiol.* 2005;288(6):R1649-R1663.
 156. Iranmanesh A, South S, Liem AY, *et al.* Unequal impact of age, percentage body fat, and serum testosterone concentrations on the somatotrophic, IGF-I, and IGF-binding protein responses to a three-day intravenous growth hormone-releasing hormone pulsatile infusion in men. *Eur J Endocrinol.* 1998;139(1):59-71.
 157. Gentili A, Mulligan T, Godschalk M, *et al.* Unequal impact of short-term testosterone repletion on the somatotrophic axis of young and older men. *J Clin Endocrinol Metab.* 2002;87(2):825-834.
 158. Moiseeva V, Cisneros A, Sica V, *et al.* Senescence atlas reveals an aged-like inflamed niche that blunts muscle regeneration. *Nature.* 2022;613(7942):169-178.
 159. Anghel L, Baroiu L, Popazu CR, *et al.* Benefits and adverse events of melatonin use in the elderly (review). *Exp Ther Med.* 2022;23(3):219.
 160. Kondratov RV, Kondratova AA, Gorbacheva VY, Vykhovanets OV, Antoch MP. Early aging and age-related pathologies in mice deficient in BMAL1, the core component of the circadian clock. *Genes Dev.* 2006;20(14):1868-1873.
 161. Antoch MP, Gorbacheva VY, Vykhovanets O, *et al.* Disruption of the circadian clock due to the Clock mutation has discrete effects on aging and carcinogenesis. *Cell Cycle.* 2008;7(9):1197-1204.
 162. Dubrovsky YV, Samsa WE, Kondratov RV. Deficiency of circadian protein CLOCK reduces lifespan and increases age-related cataract development in mice. *Aging (Albany NY).* 2010;2(12):936-944.
 163. Carroll JE, Irwin MR, Levine M, *et al.* Epigenetic aging and immune senescence in women with insomnia symptoms: findings from the women's health initiative study. *Biol Psychiatry.* 2017;81(2):136-144.
 164. Carroll JE, Olmstead R, Cole SW, Breen EC, Arevalo JM, Irwin MR. Remission of insomnia in older adults treated with cognitive behavioral therapy for insomnia (CBT-I) reduces p16(INK4a) gene expression in peripheral blood: secondary outcome analysis from a randomized clinical trial. *Geroscience.* 2023;45(4):2325-2335.
 165. Carreras A, Zhang SX, Peris E, *et al.* Chronic sleep fragmentation induces endothelial dysfunction and structural vascular changes in mice. *Sleep.* 2014;37(11):1817-1824.
 166. Ezzat S, Asa SL, Couldwell WT, *et al.* The prevalence of pituitary adenomas: a systematic review. *Cancer.* 2004;101(3):613-619.
 167. Kuilman T, Michaloglou C, Vredeveld LC, *et al.* Oncogene-induced senescence relayed by an interleukin-dependent inflammatory network. *Cell.* 2008;133(6):1019-1031.
 168. Manojlovic-Gacic E, Skender-Gazibara M, Popovic V, *et al.* Oncogene-induced senescence in pituitary adenomas--an immunohistochemical study. *Endocr Pathol.* 2016;27(1):1-11.
 169. Hansen KR, Knowlton NS, Thyer AC, Charleston JS, Soules MR, Klein NA. A new model of reproductive aging: the decline in ovarian non-growing follicle number from birth to menopause. *Hum Reprod.* 2008;23(3):699-708.
 170. El Khoudary SR, Aggarwal B, Beckie TM, *et al.* Menopause transition and cardiovascular disease risk: implications for timing of early prevention: a scientific statement from the American Heart Association. *Circulation.* 2020;142(25):e506-e532.
 171. Alberico HC, Woods DC. Role of granulosa cells in the aging ovarian landscape: a focus on mitochondrial and metabolic function. *Front Physiol.* 2021;12:800739.
 172. Secomandi L, Borghesan M, Velarde M, Demaria M. The role of cellular senescence in female reproductive aging and the potential for senotherapeutic interventions. *Hum Reprod Update.* 2022;28(2):172-189.
 173. Krishnamurthy J, Torrice C, Ramsey MR, *et al.* Ink4a/Arf expression is a biomarker of aging. *J Clin Invest.* 2004;114(9):1299-1307.
 174. Merz SE, Klopffleisch R, Breithaupt A, Gruber AD. Aging and senescence in canine testes. *Vet Pathol.* 2019;56(5):715-724.
 175. Kusama K, Yamauchi N, Yoshida K, Azumi M, Yoshie M, Tamura K. Senolytic treatment modulates decidualization in human endometrial stromal cells. *Biochem Biophys Res Commun.* 2021;571:174-180.
 176. Liu F, Wan Q, Liu P, Miao D, Dai X, Chen L. Loss of p16 does not protect against premature ovarian insufficiency caused by alkylating agents. *BMC Pregnancy Childbirth.* 2023;23(1):151.
 177. Nair KS, Rizza RA, O'Brien P, *et al.* DHEA in elderly women and DHEA or testosterone in elderly men. *N Engl J Med.* 2006;355(16):1647-1659.

178. Enomoto M, Adachi H, Fukami A, *et al.* Serum dehydroepiandrosterone sulfate levels predict longevity in men: 27-year follow-up study in a community-based cohort (Tanushimaru study). *J Am Geriatr Soc.* 2008;56(6):994-998.
179. Wierman ME, Kiseljick-Vassiliades K. Should dehydroepiandrosterone be administered to women? *J Clin Endocrinol Metab.* 2022;107(6):1679-1685.
180. Tatomir A, Micu C, Crivii C. The impact of stress and glucocorticoids on memory. *Clujul Med.* 2014;87(1):3-6.
181. Guan R, Yang C, Zhang J, Wang J, Chen R, Su P. Dehydroepiandrosterone alleviates hypoxia-induced learning and memory dysfunction by maintaining synaptic homeostasis. *CNS Neurosci Ther.* 2022;28(9):1339-1350.
182. Kau MM, Chen JJ, Wang SW, Cho WL, Wang PS. Age-related impairment of aldosterone secretion in zona glomerulosa cells of ovariectomized rats. *J Invest Med.* 1999;47(8):425-432.
183. Gao X, Li F, Liu B, Wang Y, Wang Y, Zhou H. Cellular senescence in adrenocortical biology and its disorders. *Cells.* 2021;10(12):3474.
184. De Stefano MA, Porcelli T, Ambrosio R, *et al.* Type 2 deiodinase is expressed in anaplastic thyroid carcinoma and its inhibition causes cell senescence. *Endocr Relat Cancer.* 2023;30(5):e230016.
185. Percheron G, Hogrel JY, Denot-Ledunois S, *et al.* Effect of 1-year oral administration of dehydroepiandrosterone to 60- to 80-year-old individuals on muscle function and cross-sectional area: a double-blind placebo-controlled trial. *Arch Intern Med.* 2003;163(6):720-727.
186. Bowers J, Terrien J, Clerget-Froidevaux MS, *et al.* Thyroid hormone signaling and homeostasis during aging. *Endocr Rev.* 2013;34(4):556-589.
187. Franceschi C, Ostan R, Mariotti S, Monti D, Vitale G. The aging thyroid: a reappraisal within the geroscience integrated perspective. *Endocr Rev.* 2019;40(5):1250-1270.
188. Buffenstein R, Pinto M. Endocrine function in naturally long-living small mammals. *Mol Cell Endocrinol.* 2009;299(1):101-111.
189. Zambrano A, García-Carpizo V, Gallardo ME, *et al.* The thyroid hormone receptor β induces DNA damage and premature senescence. *J Cell Biol.* 2014;204(1):129-146.
190. Chini CCS, Peclat TR, Warner GM, *et al.* CD38 ecto-enzyme in immune cells is induced during aging and regulates NAD(+) and NMN levels. *Nat Metab.* 2020;2(11):1284-1304.
191. Lagnado A, Leslie J, Ruchaud-Sparagano MH, *et al.* Neutrophils induce paracrine telomere dysfunction and senescence in ROS-dependent manner. *EMBO J.* 2021;40(9):e106048.
192. Kammori M, Nakamura K, Kawahara M, Mimura Y, Kaminishi M, Takubo K. Telomere shortening with aging in human thyroid and parathyroid tissue. *Exp Gerontol.* 2002;37(4):513-521.
193. Bates JN, Kohn TP, Pastuszak AW. Effect of thyroid hormone derangements on sexual function in men and women. *Sex Med Rev.* 2020;8(2):217-230.
194. Meng Z, Liu M, Zhang Q, *et al.* Gender and age impact on the association between thyroid-stimulating hormone and serum lipids. *Medicine (Baltimore).* 2015;94(49):e2186.
195. Riis J, Pedersen KM, Danielsen MB, *et al.* Long-term iodine nutrition is associated with longevity in older adults: a 20 years' follow-up of the Randers-Skagen study. *Br J Nutr.* 2021;125(3):260-265.
196. Tolu F, Palermo M, Dore MP, *et al.* Association of endemic goitre and exceptional longevity in Sardinia: evidence from an ecological study. *Eur J Ageing.* 2019;16(4):405-414.
197. Jansen SW, Akintola AA, Roelfsema F, *et al.* Human longevity is characterised by high thyroid stimulating hormone secretion without altered energy metabolism. *Sci Rep.* 2015;5(1):11525.
198. Maniakas A, Dadu R, Busaidy NL, *et al.* Evaluation of overall survival in patients with anaplastic thyroid carcinoma, 2000–2019. *JAMA Oncol.* 2020;6(9):1397-1404.
199. Prasanna PG, Citrin DE, Hildesheim J, *et al.* Therapy-Induced senescence: opportunities to improve anticancer therapy. *J Natl Cancer Inst.* 2021;113(10):1285-1298.
200. Guida JL, Agurs-Collins T, Ahles TA, *et al.* Strategies to prevent or remediate cancer and treatment-related aging. *J Natl Cancer Inst.* 2021;113(2):112-122.
201. Chalcraft JR, Cardinal LM, Wechsler PJ, *et al.* Vitamin D synthesis following a single bout of sun exposure in older and younger men and women. *Nutrients.* 2020;12(8):2237.
202. Wacker M, Holick MF. Sunlight and vitamin D: a global perspective for health. *Dermatoendocrinol.* 2013;5(1):51-108.
203. Campisi J. The role of cellular senescence in skin aging. *J Invest Dermatol Symp Proc.* 1998;3(1):1-5.
204. Wyles SP, Tchkonja T, Kirkland JL. Targeting cellular senescence for age-related diseases: path to clinical translation. *Plast Reconstr Surg.* 2022;150:20s-26s.
205. Gonçalves de Carvalho CM, Ribeiro SM. Aging, low-grade systemic inflammation and vitamin D: a mini-review. *Eur J Clin Nutr.* 2017;71(4):434-440.
206. de Jongh RT, van Schoor NM, Lips P. Changes in vitamin D endocrinology during aging in adults. *Mol Cell Endocrinol.* 2017;453:144-150.
207. Hill TR, Granic A, Aspray TJ. Vitamin D and ageing. *Subcell Biochem.* 2018;90:191-220.
208. Gallagher JC. Vitamin D and aging. *Endocrinol Metab Clin North Am.* 2013;42(2):319-332.
209. Amrein K, Scherkl M, Hoffmann M, *et al.* Vitamin D deficiency 2.0: an update on the current status worldwide. *Eur J Clin Nutr.* 2020;74(11):1498-1513.
210. Vranic L, Mikolašević I, Milić S. Vitamin D deficiency: consequence or cause of obesity? *Medicina (Kaunas).* 2019;55(9):541.
211. Lips P, van Schoor NM. The effect of vitamin D on bone and osteoporosis. *Best Pract Res Clin Endocrinol Metab.* 2011;25(4):585-591.
212. Richards JB, Valdes AM, Gardner JP, *et al.* Higher serum vitamin D concentrations are associated with longer leukocyte telomere length in women. *Am J Clin Nutr.* 2007;86(5):1420-1425.
213. Chen L, Yang R, Qiao W, *et al.* 1,25-Dihydroxyvitamin d exerts an antiaging role by activation of Nrf2-antioxidant signaling and inactivation of p16/p53-senescence signaling. *Aging Cell.* 2019;18(3):e12951.
214. Barbouti A, Evangelou K, Pateras IS, *et al.* In situ evidence of cellular senescence in thymic epithelial cells (TECs) during human thymic involution. *Mech Ageing Dev.* 2019;177:88-90.
215. Pan XH, Lin QK, Yao X, *et al.* Umbilical cord mesenchymal stem cells protect thymus structure and function in aged C57 mice by downregulating aging-related genes and upregulating autophagy- and anti-oxidative stress-related genes. *Aging (Albany NY).* 2020;12(17):16899-16920.
216. Palmer S, Albergante L, Blackburn CC, Newman TJ. Thymic involution and rising disease incidence with age. *Proc Natl Acad Sci U S A.* 2018;115(8):1883-1888.
217. Aw D, Silva AB, Maddick M, von Zglinicki T, Palmer DB. Architectural changes in the thymus of aging mice. *Aging Cell.* 2008;7(2):158-167.
218. Thomas R, Wang W, Su D-M. Contributions of age-related thymic involution to immunosenescence and inflammaging. *Immun Ageing.* 2020;17(1):2.
219. van der Loo B, Labugger R, Skepper JN, *et al.* Enhanced peroxynitrite formation is associated with vascular aging. *J Exp Med.* 2000;192(12):1731-1744.
220. Sato I, Kaji K, Morita I, Nagao M, Murota S-I. Augmentation of endothelin-1, prostacyclin and thromboxane A2 secretion associated with in vitro ageing in cultured human umbilical vein endothelial cells. *Mech Ageing Dev.* 1993;71(1-2):73-84.
221. Comi P, Chiamonte R, Maier JA. Senescence-dependent regulation of type 1 plasminogen activator inhibitor in human vascular endothelial cells. *Exp Cell Res.* 1995;219(1):304-308.

222. Garfinkel S, Brown S, Wessendorf JH, Maciag T. Post-transcriptional regulation of interleukin 1 alpha in various strains of young and senescent human umbilical vein endothelial cells. *Proc Natl Acad Sci U S A*. 1994;91(4):1559-1563.
223. Minamino T, Miyauchi H, Yoshida T, Ishida Y, Yoshida H, Komuro I. Endothelial cell senescence in human atherosclerosis: role of telomere in endothelial dysfunction. *Circulation*. 2002;105(13):1541-1544.
224. Yokoyama M, Shimizu I, Nagasawa A, *et al*. P53 plays a crucial role in endothelial dysfunction associated with hyperglycemia and ischemia. *J Mol Cell Cardiol*. 2019;129:105-117.
225. Yokoyama M, Okada S, Nakagomi A, *et al*. Inhibition of endothelial p53 improves metabolic abnormalities related to dietary obesity. *Cell Rep*. 2014;7(5):1691-1703.
226. Suda M, Shimizu I, Katsumi G, *et al*. Senolytic vaccination improves normal and pathological age-related phenotypes and increases lifespan in progeroid mice. *Nat Aging*. 2021;1(12):1117-1126.
227. Kunieda T, Minamino T, Nishi J, *et al*. Angiotensin II induces premature senescence of vascular smooth muscle cells and accelerates the development of atherosclerosis via a p21-dependent pathway. *Circulation*. 2006;114(9):953-960.
228. Faget DV, Ren Q, Stewart SA. Unmasking senescence: context-dependent effects of SASP in cancer. *Nat Rev Cancer*. 2019;19(8):439-453.
229. Biron-Shental T, Kidron D, Sukenik-Halevy R, *et al*. TERC telomerase subunit gene copy number in placentas from pregnancies complicated with intrauterine growth restriction. *Early Hum Dev*. 2011;87(2):73-75.
230. Heazell AE, Sharp AN, Baker PN, Crocker IP. Intra-uterine growth restriction is associated with increased apoptosis and altered expression of proteins in the p53 pathway in villous trophoblast. *Apoptosis*. 2011;16(2):135-144.
231. Xu M, Pirtskhalava T, Farr JN, *et al*. Senolytics improve physical function and increase lifespan in old age. *Nat Med*. 2018;24(8):1246-1256.
232. Cubro H, Nath KA, Suvakov S, *et al*. Mechanisms of vascular dysfunction in the interleukin-10-deficient murine model of preeclampsia indicate nitric oxide dysregulation. *Kidney Int*. 2021;99(3):646-656.
233. Suvakov S, Cubro H, White WM, *et al*. Targeting senescence improves angiogenic potential of adipose-derived mesenchymal stem cells in patients with preeclampsia. *Biol Sex Differ*. 2019;10(1):49.
234. Suvakov S, Ghamrawi R, Cubro H, *et al*. Epigenetic and senescence markers indicate an accelerated ageing-like state in women with preeclamptic pregnancies. *EBioMedicine*. 2021;70:103536.
235. Tchakonia T, Kirkland JL. Aging, cell senescence, and chronic disease: emerging therapeutic strategies. *JAMA*. 2018;320(13):1319-1320.
236. Zhang X, Englund DA, Aversa Z, Jachim SK, White TA, LeBrasseur NK. Exercise counters the age-related accumulation of senescent cells. *Exerc Sport Sci Rev*. 2022;50(4):213-221.
237. Wiley CD, Campisi J. The metabolic roots of senescence: mechanisms and opportunities for intervention. *Nat Metab*. 2021;3(10):1290-1301.
238. Miller RA, Buehner G, Chang Y, Harper JM, Sigler R, Smith-Wheelock M. Methionine-deficient diet extends mouse lifespan, slows immune and lens aging, alters glucose, T4, IGF-I and insulin levels, and increases hepatocyte MIF levels and stress resistance. *Aging Cell*. 2005;4(3):119-125.
239. Newman JC, Covarrubias AJ, Zhao M, *et al*. Ketogenic diet reduces midlife mortality and improves memory in aging mice. *Cell Metab*. 2017;26(3):547-557.e8.
240. Roberts MN, Wallace MA, Tomilov AA, *et al*. A ketogenic diet extends longevity and healthspan in adult mice. *Cell Metab*. 2017;26(3):539-546.e5.
241. Nilsson MI, Bourgeois JM, Nederveen JP, *et al*. Lifelong aerobic exercise protects against inflammaging and cancer. *PLoS One*. 2019;14(1):e0210863.
242. Garcia-Valles R, Gomez-Cabrera MC, Rodriguez-Mañás L, *et al*. Life-long spontaneous exercise does not prolong lifespan but improves health span in mice. *Longev Healthspan*. 2013;2(1):14.
243. Schafer MJ, Mazula DL, Brown AK, *et al*. Late-life time-restricted feeding and exercise differentially alter healthspan in obesity. *Aging Cell*. 2019;18(4):e12966.
244. Wen J, Bao M, Tang M, He X, Yao X, Li L. Low magnitude vibration alleviates age-related bone loss by inhibiting cell senescence of osteogenic cells in naturally senescent rats. *Aging (Albany NY)*. 2021;13(8):12031-12045.
245. Simpson RJ, Cosgrove C, Chee MM, *et al*. Senescent phenotypes and telomere lengths of peripheral blood T-cells mobilized by acute exercise in humans. *Exerc Immunol Rev*. 2010;16:40-55.
246. Walton RG, Kosmac K, Mula J, *et al*. Human skeletal muscle macrophages increase following cycle training and are associated with adaptations that may facilitate growth. *Sci Rep*. 2019;9(1):969.
247. Nielsen HB, Secher NH, Christensen NJ, Pedersen BK. Lymphocytes and NK cell activity during repeated bouts of maximal exercise. *Am J Physiol*. 1996;271(1):R222-R227.
248. Krizhanovsky V, Yon M, Dickins RA, *et al*. Senescence of activated stellate cells limits liver fibrosis. *Cell*. 2008;134(4):657-667.
249. Shimizu I, Yoshida Y, Katsuno T, *et al*. p53-induced adipose tissue inflammation is critically involved in the development of insulin resistance in heart failure. *Cell Metab*. 2012;15(1):51-64.
250. Krimpenfort P, Quon KC, Mooi WJ, Loonstra A, Berns A. Loss of p16Ink4a confers susceptibility to metastatic melanoma in mice. *Nature*. 2001;413(6851):83-86.
251. Martín-Caballero J, Flores JM, García-Palencia P, Serrano M. Tumor susceptibility of p21(Waf1/Cip1)-deficient mice. *Cancer Res*. 2001;61(16):6234-6238.
252. Sharpless NE, Bardeesy N, Lee KH, *et al*. Loss of p16Ink4a with retention of p19Arf predisposes mice to tumorigenesis. *Nature*. 2001;413(6851):86-91.
253. Takeuchi S, Takahashi A, Motoi N, *et al*. Intrinsic cooperation between p16INK4a and p21Waf1/Cip1 in the onset of cellular senescence and tumor suppression in vivo. *Cancer Res*. 2010;70(22):9381-9390.
254. Di Micco R, Krizhanovsky V, Baker D, d'Adda di Fagnaga F. Cellular senescence in ageing: from mechanisms to therapeutic opportunities. *Nat Rev Mol Cell Biol*. 2021;22(2):75-95.
255. Wang B, Wang L, Gasek NS, *et al*. An inducible p21-Cre mouse model to monitor and manipulate p21-highly-expressing senescent cells in vivo. *Nat Aging*. 2021;1(10):962-973.
256. Wang L, Wang B, Gasek NS, *et al*. Targeting p21(Cip1) highly expressing cells in adipose tissue alleviates insulin resistance in obesity. *Cell Metab*. 2022;34(1):186.
257. Ramos FJ, Chen SC, Garelick MG, *et al*. Rapamycin reverses elevated mTORC1 signaling in lamin A/C-deficient mice, rescues cardiac and skeletal muscle function, and extends survival. *Sci Transl Med*. 2012;4(144):144ra103.
258. Daneshgar N, Rabinovitch PS, Dai DF. TOR signaling pathway in cardiac aging and heart failure. *Biomolecules*. 2021;11(2):168.
259. Laberge RM, Sun Y, Orjalo AV, *et al*. MTOR regulates the pro-tumorigenic senescence-associated secretory phenotype by promoting IL1A translation. *Nat Cell Biol*. 2015;17(8):1049-1061.
260. Spilman P, Podlaskaya N, Hart MJ, *et al*. Inhibition of mTOR by rapamycin abolishes cognitive deficits and reduces amyloid-beta levels in a mouse model of Alzheimer's disease. *PLoS One*. 2010;5(4):e9979.
261. Hurez V, Dao V, Liu A, *et al*. Chronic mTOR inhibition in mice with rapamycin alters T, B, myeloid, and innate lymphoid cells and gut flora and prolongs life of immune-deficient mice. *Aging Cell*. 2015;14(6):945-956.
262. Walters HE, Cox LS. mTORC inhibitors as broad-spectrum therapeutics for age-related diseases. *Int J Mol Sci*. 2018;19(8):2325.
263. Kandhaya-Pillai R, Yang X, Tchakonia T, Martin GM, Kirkland JL, Oshima J. TNF- α /IFN- γ synergy amplifies senescence-associated inflammation and SARS-CoV-2 receptor expression via hyper-activated JAK/STAT1. *Aging Cell*. 2022;21(6):e13646.

264. Xu M, Tchkonja T, Ding H, *et al.* JAK inhibition alleviates the cellular senescence-associated secretory phenotype and frailty in old age. *Proc Natl Acad Sci U S A.* 2015;112(46):E6301-E6310.
265. Addinsall AB, Cacciani N, Akkad H, *et al.* JAK/STAT inhibition augments soleus muscle function in a rat model of critical illness myopathy via regulation of complement C3/3R. *J Physiol.* 2021;599(11):2869-2886.
266. Verstovsek S, Kantarjian H, Mesa RA, *et al.* Safety and efficacy of INCB018424, a JAK1 and JAK2 inhibitor, in myelofibrosis. *N Engl J Med.* 2010;363(12):1117-1127.
267. Moiseeva O, Deschênes-Simard X, St-Germain E, *et al.* Metformin inhibits the senescence-associated secretory phenotype by interfering with IKK/NF- κ B activation. *Aging Cell.* 2013;12(3):489-498.
268. Effect of intensive blood-glucose control with metformin on complications in overweight patients with type 2 diabetes (UKPDS 34). UK Prospective Diabetes Study (UKPDS) group. *Lancet.* 1998;352(9131):854-865.
269. Barzilai N, Crandall JP, Kritchevsky SB, Espeland MA. Metformin as a tool to target aging. *Cell Metab.* 2016;23(6):1060-1065.
270. Huffman DM, Justice JN, Stout MB, Kirkland JL, Barzilai N, Austad SN. Evaluating health span in preclinical models of aging and disease: guidelines, challenges, and opportunities for geroscience. *J Gerontol A Biol Sci Med Sci.* 2016;71(11):1395-1406.
271. Bannister CA, Holden SE, Jenkins-Jones S, *et al.* Can people with type 2 diabetes live longer than those without? A comparison of mortality in people initiated with metformin or sulphonylurea monotherapy and matched, non-diabetic controls. *Diabetes Obes Metab.* 2014;16(11):1165-1173.
272. American Federation for Aging Research. Targeting the biology of aging, ushering a new era of interventions. Accessed April 5, 2024. <https://www.afar.org/tame-trial>
273. Justice JN, Ferrucci L, Newman AB, *et al.* A framework for selection of blood-based biomarkers for geroscience-guided clinical trials: report from the TAME biomarkers workgroup. *Geroscience.* 2018;40(5-6):419-436.
274. Quarles E, Basisty N, Chiao YA, *et al.* Rapamycin persistently improves cardiac function in aged, male and female mice, even following cessation of treatment. *Aging Cell.* 2020;19(2):e13086.
275. Baker DJ, Wijshake T, Tchkonja T, *et al.* Clearance of p16lnk4a-positive senescent cells delays ageing-associated disorders. *Nature.* 2011;479(7372):232-236.
276. Pajvani UB, Trujillo ME, Combs TP, *et al.* Fat apoptosis through targeted activation of caspase 8: a new mouse model of inducible and reversible lipodystrophy. *Nat Med.* 2005;11(7):797-803.
277. Kirkland JL, Tchkonja T. Senolytic drugs: from discovery to translation. *J Intern Med.* 2020;288(5):518-536.
278. Wissler Gerdes EO, Zhu Y, Tchkonja T, Kirkland JL. Discovery, development, and future application of senolytics: theories and predictions. *FEBS J.* 2020;287(12):2418-2427.
279. Mikawa R, Suzuki Y, Baskoro H, *et al.* Elimination of p19(ARF)-expressing cells protects against pulmonary emphysema in mice. *Aging Cell.* 2018;17(5):e12827.
280. Wang E. Senescent human fibroblasts resist programmed cell death, and failure to suppress bcl2 is involved. *Cancer Res.* 1995;55(11):2284-2292.
281. Passos JF, Saretzki G, Ahmed S, *et al.* Mitochondrial dysfunction accounts for the stochastic heterogeneity in telomere-dependent senescence. *PLoS Biol.* 2007;5(5):e110.
282. Wang Y, Chang J, Liu X, *et al.* Discovery of piperlongumine as a potential novel lead for the development of senolytic agents. *Aging (Albany NY).* 2016;8(11):2915-2926.
283. Kirkland JL, Tchkonja T. Cellular senescence: a translational perspective. *EBioMedicine.* 2017;21:21-28.
284. Passos JF, Nelson G, Wang C, *et al.* Feedback between p21 and reactive oxygen production is necessary for cell senescence. *Mol Syst Biol.* 2010;6(1):347.
285. Chang Q, Jorgensen C, Pawson T, Hedley DW. Effects of dasatinib on EphA2 receptor tyrosine kinase activity and downstream signaling in pancreatic cancer. *Br J Cancer.* 2008;99(7):1074-1082.
286. Xi HQ, Wu XS, Wei B, Chen L. Eph receptors and ephrins as targets for cancer therapy. *J Cell Mol Med.* 2012;16(12):2894-2909.
287. Chang J, Wang Y, Shao L, *et al.* Clearance of senescent cells by ABT263 rejuvenates aged hematopoietic stem cells in mice. *Nat Med.* 2016;22(1):78-83.
288. Zhu Y, Tchkonja T, Fuhrmann-Stroissnigg H, *et al.* Identification of a novel senolytic agent, navitoclax, targeting the Bcl-2 family of anti-apoptotic factors. *Aging Cell.* 2016;15(3):428-435.
289. Zhu Y, Doornebal EJ, Pirtskhalava T, *et al.* New agents that target senescent cells: the flavone, fisetin, and the BCL-X(L) inhibitors, A1331852 and A1155463. *Aging (Albany NY).* 2017;9(3):955-963.
290. Yousefzadeh MJ, Zhu Y, McGowan SJ, *et al.* Fisetin is a senotherapeutic that extends health and lifespan. *EBioMedicine.* 2018;36:18-28.
291. Xu Q, Fu Q, Li Z, *et al.* The flavonoid procyanidin C1 has senotherapeutic activity and increases lifespan in mice. *Nat Metab.* 2021;3(12):1706-1726.
292. Baar MP, Brandt RMC, Putavet DA, *et al.* Targeted apoptosis of senescent cells restores tissue homeostasis in response to chemotoxicity and aging. *Cell.* 2017;169(1):132-147.e16.
293. Guerrero A, Herranz N, Sun B, *et al.* Cardiac glycosides are broad-spectrum senolytics. *Nat Metab.* 2019;1(11):1074-1088.
294. Triana-Martínez F, Picallos-Rabina P, Da Silva-Álvarez S, *et al.* Identification and characterization of Cardiac Glycosides as senolytic compounds. *Nat Commun.* 2019;10(1):4731.
295. Johmura Y, Yamanaka T, Omori S, *et al.* Senolysis by glutaminolysis inhibition ameliorates various age-associated disorders. *Science.* 2021;371(6526):265-270.
296. Suda M, Shimizu I, Katsuomi G, *et al.* Glycoprotein nonmetastatic melanoma protein B regulates lysosomal integrity and lifespan of senescent cells. *Sci Rep.* 2022;12(1):6522.
297. Yoshida S, Nakagami H, Hayashi H, *et al.* The CD153 vaccine is a senotherapeutic option for preventing the accumulation of senescent T cells in mice. *Nat Commun.* 2020;11(1):2482.
298. Amor C, Feucht J, Leibold J, *et al.* Senolytic CAR T cells reverse senescence-associated pathologies. *Nature.* 2020;583(7814):127-132.
299. Samakkarnthai P, Saul D, Zhang L, *et al.* In vitro and in vivo effects of zoledronic acid on senescence and senescence-associated secretory phenotype markers. *Aging (Albany NY).* 2023;15(9):3331-3355.
300. Takahashi K, Yamanaka S. Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors. *Cell.* 2006;126(4):663-676.
301. Ocampo A, Reddy P, Martínez-Redondo P, *et al.* In vivo amelioration of age-associated hallmarks by partial reprogramming. *Cell.* 2016;167(7):1719-1733.e12.
302. Xu M, Bradley EW, Weivoda MM, *et al.* Transplanted senescent cells induce an osteoarthritis-like condition in mice. *J Gerontol A Biol Sci Med Sci.* 2017;72(6):780-785.
303. Gonzalez-Meljem JM, Martínez-Barbera JP. Adamantinomatous craniopharyngioma as a model to understand paracrine and senescence-induced tumorigenesis. *Cell Mol Life Sci.* 2021;78(10):4521-4544.
304. Gonzalez-Meljem JM, Haston S, Carreno G, *et al.* Stem cell senescence drives age-attenuated induction of pituitary tumours in mouse models of paediatric craniopharyngioma. *Nat Commun.* 2017;8(1):1819.
305. Justice JN, Nambiar AM, Tchkonja T, *et al.* Senolytics in idiopathic pulmonary fibrosis: results from a first-in-human, open-label, pilot study. *EBioMedicine.* 2019;40:554-563.
306. Takasugi M, Yoshida Y, Hara E, Ohtani N. The role of cellular senescence and SASP in tumour microenvironment. *FEBS J.* 2023;290(5):1348-1361.
307. Rao SG, Jackson JG. SASP: tumor suppressor or promoter? Yes!. *Trends Cancer.* 2016;2(11):676-687.